

The Final Fifty for Datasets 51–100, with Ten in Reserve

Michael Volk

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The database holds 50 schematized and L0–L4-verified *Saccharomyces cerevisiae* datasets, of which 36 are served and 14 are built and waiting on the next full graph rebuild. Table 1 is the final selection for datasets 51 to 100: the fifty recommended builds in rank order, then ten extra in rank order so that a row which proves unreachable has a named replacement. Every figure the ranking used sits in that table beside the reason the row is there. Table 2 gives each row’s citation, link, data location and the reason for its position; Table 3 names the joins each row opens against a built dataset or another row; Table 4 records what was considered and dropped; Sec. 6 closes with the summary statistics and the two review points from the previous version.

The order is a rule, not a taste (Sec. 1): the rows a planned yeast Perturb-seq is designed against come first, then the rows that put two molecular layers on one joinable axis, then scale; inside a band, tier, then measurements. The fifty are exactly the first two bands, 30 rows and 20. The ten extra are the last three molecular-layers rows and the first seven scale rows. The 108 rows ranked below them are kept in the generating script and its JSON and are not printed here.

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1 The selection rule ✗

1.1 Sequence reconstruction is the gate, not a preference ✗

A phenotype is only trainable against a genotype that can be written down. Every row in Table 1 therefore names a **sequence basis**: the route by which the total genomic content of one strain would be reconstructed. The eleven bases in use are

- **S288C-KO** – the reference genome minus one cataloged ORF (BY4741, BY4742, BY4743). Thirteen of the sixty in Table 1 sit here, second only to reference-only.
- **S288C-KO/het** – a heterozygous diploid deletion, so a dosage edit rather than a null.
- **S288C+guide** – the genome is unedited and the perturbation is a designed cassette plus a guide target. Regulatory, not sequence-level.
- **S288C+designed-edit** – a designed single-nucleotide or small-indel library. Designed, and not verified per strain.
- **S288C+tag** – the reference plus a designed tag, degron or promoter cassette.
- **S288C+ORF-plasmid** – the reference plus a barcoded ORF on a known plasmid.
- **S288C+reporter-locus** – the reference plus a designed regulatory sequence driving a reporter, which is what every massively parallel reporter assay perturbs.
- **isolate-WGS** – a published assembly or variant call set per isolate.
- **segregant-WGS** – genotype calls or whole-genome sequence per progeny.
- **engineered-chassis** – a named production strain plus heterologous cassettes.
- **reference-only** – wild type, where the environment carries the perturbation.

What fails the gate is a strain whose genome is unknown even in principle. Genome shuffling and adaptive laboratory evolution without resequencing both land there: the selected progeny differ from their parents at unmapped positions, so there is no genotype to map a phenotype from. Those are dropped in Table 4 rather than ranked low, since no amount of phenotypic value substitutes for a missing genotype.

The **S288C+designed-edit** and **S288C+guide** bases carry a weaker claim than the others and the distinction is worth holding. A designed variant library states what each member was built to be, not what it turned out to be, so its genotypes are designed rather than realized. Ingesting one commits to storing the design and a verification status, not a verified sequence.

1.2 Band comes before tier, and why ✗

Scale is the right default ordering and it is the wrong first question when a specific campaign is being planned. The largest row in this table, de Boer 2020 at 10^8 promoter sequences, tells a Perturb-seq design nothing; a 61,094-guide CRISPR interference library with a released per-guide matrix tells it what guides to order. Each row therefore carries a **band**, applied before tier and before scale:

- **perturb-seq**, 30 rows. A CRISPR interference or activation library whose per-guide matrix is released, a barcoded genotype set read out per cell, a designed single-gene perturbation crossed with a transcriptome, or the strain and noise measurements a pooled per-cell campaign depends on.
- **molecular layers**, 23 rows. Two or more of transcript, translation, protein, phosphosite, metabolite, flux and turnover, joinable on one axis, either inside one study or against a supported dataset. The join is per *strain* when both layers are measured on the same genotypes, and per *gene* when one layer is a single genome-wide coefficient such as a half-life.
- **scale**, 115 rows. Everything else, ordered as before.

The second band was **metabolism x expression** and is widened, because transcript against protein is the same join as expression against flux, and because the rows that say *why* transcript and protein disagree, translation efficiency and turnover, had no band to sit in at all. Its membership is unchanged apart from the rows added with it.

Within a band the order is tier, then measurements. The cost is visible and is not hidden: the two bands total 53 rows, so they fill the fifty and three of them become the first three of the ten extra, and no scale row reaches the fifty at all. de Boer 2020 sits at 54 and Lee 2014 at 56. Both are the largest things in the ranking by instance count and both are in the extra ten, which is the deliberate consequence of ordering by what a row combines with rather than by how many numbers it holds. A reader who wants scale back reads rows 54 to 60 first.

1.3 Tier, and what rows are ranked on ✗

Tier is assigned from stated bars rather than judgment:

- **Tier 1** clears all three of $\geq 10^3$ genotypes, $\geq 10^4$ instances, and a clean sequence basis, and lands in one of the nine classes.
- **Tier 2** clears two of the three, or is the only dataset covering a requested class, or directly de-risks the Perturb-seq proposal.
- **Tier 3** is a modality or reference backbone the substrate needs without itself carrying a large genotype axis.
- **Tier 4** is real but small, coarsely labeled, or expensive to extract.

The bars apply to what a row contains rather than to its reputation. Piotrowski 2017 has the largest compound library in yeast at 13,524 compounds and still fails Tier 1, because 157 strains is not a genotype axis. Turco 2023 has the widest environment axis at 7,536 and fails for a different reason: it is a meta-aggregation, so its instance count re-serves primary screens, several of them already built, and the net-new fraction is unknown until it is de-duplicated against the rest of this table.

Within a tier, rows sort by **measurements**, meaning instances times phenotype dimensionality. Instances alone is the wrong statistic and gets the order visibly wrong: Muenzner 2024 is 796 instances, which ranks it below a mid-sized fitness screen, but each instance is a proteome, so it is roughly 1.6×10^6 numbers. This is the same normalization the supported-datasets table already applies through its gzip-signal column.

1.4 Where the lines fall, and what they cost ✗

The long-run cut at 150 is arithmetic, $200 - 50$, not a quality threshold. Rows on both sides of it are close, so it should be read as a budget rather than a verdict.

The two nearer lines are what a build week is planned against. Wave 1 is rows 1–50 and wave 2 rows 51–70; a wave-2 row is promoted the moment a wave-1 row turns out to have no recoverable per-strain data, which has already happened once to Lee 2014. Wave 1 is 30 Perturb-seq rows and 20 molecular-layer rows, with no scale row in it, and wave 2 is three molecular-layer rows followed by scale, so the two waves differ in kind and not only in rank.

A dataset named explicitly in the scoping request is pinned into the recommended set even if it ranks below the cut, and the pin is reported rather than folded into the score, so the set stays auditable. No pin is currently in force: Jackson 2020, the one requested row, ranks 3 unaided.

Displacement, when a pin does bind, picks the weakest non-requested row by tier first and measurements second. The tier term is load-bearing rather than decorative: ranking on measurements alone selected Puddu 2019 for eviction, and that row is the whole-collection sequencing on which every **S288C-KO** sequence basis in this table rests. Banding has since moved Puddu to 4 on its own merits, because a pooled campaign reads a barcode and infers a genotype and this is the measurement of how often that inference is wrong.

2 The final fifty, and ten extra ✗

Table 1: The final fifty, then ten extra in rank order in case a row above proves unreachable. Every stat the ranking used is here beside the reason. *Band, tier:* *P* is the perturb-seq band, *M* molecular layers, *S* scale, applied before tier (Sec. 1); within a band the tier rule orders, then measurements. *Genotypes* and *Env* are the perturbation and condition axes. *Inst.* is genotype $\times$ environment records, $\dagger$ where it is the product of the two axes rather than a reported count and $\ddagger$ where it is an order-of-magnitude estimate. *Meas.* is instances times phenotype dimensionality, the quantity rows are ranked on. *Sequence basis* is the route to each strain’s total genomic content; a row with no route is excluded (Table 4). *Time* names the row’s time dimension where it has one; a dash means steady state or endpoint. A $\bullet$ marks a row high on a Perturb-seq axis; superscript **B** marks a row blocked on data access, **L** one with a loader in flight. Citations, links and data locations are in Table 2; joins in Table 3.

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
1	Boocock 2025 (single-cell eQTL mapping) $\bullet$ Expression / single cell; single-cell transcriptome + eQTL	P, 1	393 segregants, 27,744 cells	glucose / galactose	$5.5 \times 10^4 \dagger$	3.3×10^8	segregant-WGS	–	27,744 single cells drawn from 393 sequenced recombinants in two carbon sources, a median of 17 cells per segregant. High-dimensional on both axes at once, which almost nothing in yeast is, and the closest existing template for the proposed design.
2	Hale 2024 (CRISPRi x natural variation) $\bullet$ Natural variation; relative fitness (double-barcode seq)	P, 1	169 segregants x 1,721 genes	2 induced	$1.4 \times 10^6 \dagger$	1.4×10^6	segregant-WGS	–	Directly measures how a genetic perturbation’s effect changes with genetic background. The single best dataset for asking whether a model trained on one background transfers to another.
3	Jackson 2020 (TF-deletion single-cell atlas) $\bullet$ Expression / single cell; single-cell transcriptome	P, 1	12 genotypes (wild type + 11 TF deletions), 72 barcoded strains	11 growth conditions	$132 \dagger$	7.9×10^5	S288C-KO	–	The closest thing to a yeast Perturb-seq that exists. A transcribed barcode in the 3’ UTR of the marker cassette labels every cell with its genotype, so 72 strains, 12 genotypes at 6 independently constructed replicates, are pooled and read out per cell; every strain appears in every one of the 11 conditions. The perturbations are homozygous diploid whole-ORF deletions in a prototrophic FY4/FY5 background rather than the Giaever collection, which is what permits the minimal and nitrogen-limited media. The DNA-only barcodes of the Giaever collection are the stated reason it could not be used, and that is the design constraint any campaign here inherits.
4	Puddu 2019 (WGS of the deletion collection) Modality / backbone; structural variants, CNV, aneuploidy	P, 1	4,800 deletion strains	1	4.8×10^3	4.8×10^3	S288C-KO	–	Whole-genome sequence for every strain in the deletion collection. This is what turns the S288C-KO sequence basis from an assumption into a measurement, and it applies to roughly half the rows in this table at once.
5	N’Guessan 2025 (segregant scRNA-seq eQTL) $\bullet$ Expression / single cell; single-cell transcriptome + eQTL	P, 2	4,500 profiled segregants	1	4.5×10^3	2.7×10^7	segregant-WGS	–	Single-cell transcriptomes indexed by recombinant natural genotype rather than by an engineered knockout. The closest existing analog to the proposed Perturb-seq design, and the only one that already pairs per-cell expression with sequenced genotypes at scale.
6	Hackett 2020 (IDEA inducible-TF transcriptome time series) $\bullet$ Expression / single cell; bulk transcriptome per time point	P, 2	over 200 inducible-TF strains	estradiol induction, about eight time points	$1.6 \times 10^3 \dagger$	9.6×10^6	S288C+ designed-edit	about eight time points after induction; one record per strain and time point	Each TF is individually induced from a designed synthetic cassette and the whole transcriptome is followed over time, which is the only yeast dataset where a single-gene perturbation is crossed with a dense time axis. scYeast used it as the target for its time-series perturbation-response task (Fan 2027, Sec. 4.1.7). The instance count treats each strain-by-time-point as a record; per strain it is about 200.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
7	Hu 2007 (TF deletion expression compendium) • Expression / single cell; mRNA abundance	P, 2	269 TF deletion strains	1	269	1.6×10^6	S288C-KO	–	Transcriptomes for the regulators specifically, predating and complementing the already-built Kemmeren compendium. Deleting a transcription factor is the perturbation whose transcriptome response is most interpretable.
8	Dong 2021 (MAGIC + SAM biosensor) CRISPR library screen; SAM biosensor fluorescence (FACS)	P, 2	100,000 guides (a/i/d)	1	$1.0 \times 10^5 \ddagger$	1.0×10^5	S288C+guide	–	MAGIC re-run with a metabolite biosensor instead of a growth selection, so the label is product concentration rather than fitness. The pattern to copy for every precursor we care about, and the sibling of the already-built Lian 2019 furfural screen.
9	Momen-Roknabadi 2020 (inducible CRISPRi library) CRISPR library screen; per-guide fitness under induction	P, 2	genome-wide guide library	nutrient limitations	$9.0 \times 10^4 \ddagger$	9.0×10^4	S288C+guide	–	A second CRISPRi library on a different vector and guide-design rule. Whether a model trained on one library transfers to the other is a cheap, decisive test of guide-level overfitting. Not a Perturb-seq row: one edit per cell and a scalar fitness readout is low-dimensional on both axes, however large the library.
10	McGlinchy 2021 (genome-scale CRISPRi library) • CRISPR library screen; per-guide fitness	P, 2	61,094 guides, 10/gene	continuous culture	6.1×10^4	6.1×10^4	S288C+guide	–	Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.
11	Bao 2018 (CHAnGE single-nucleotide library) CRISPR library screen; variant fitness (pooled barcode seq)	P, 2	tens of thousands of designed variants	inhibitor selection	$6.0 \times 10^4 \ddagger$	6.0×10^4	S288C+ designed-edit	–	Single-nucleotide resolution rather than whole-gene nulls. The only genotype axis in this table finer than an ORF, and the one that matches how a real strain-design edit is specified.
12	Crook 2016 (tunable RNAi, isobutanol + 1-butanol) Tolerance / robustness; dose-graded knockdown fitness under alcohol stress	P, 2	genome-scale tunable RNAi	isobutanol, 1-butanol	$2.0 \times 10^4 \ddagger$	2.0×10^4	S288C+guide	–	Two related alcohols at graded knockdown strength. The dosage axis is the signal a binary screen throws away, and Hsp70 emerged only because of it.
13	Roy 2018 (multiplexed precision editing) CRISPR library screen; variant fitness	P, 2	thousands of designed edits	1	$1.6 \times 10^4 \ddagger$	1.6×10^4	S288C+ designed-edit	–	Genomic rather than plasmid barcodes, which removes the barcode-swapping artifact that quietly corrupts pooled fitness data. The methodological complement to CHAnGE.
14	Newman 2006 (protein noise, GFP library) • Modality / backbone; protein abundance and cell-to-cell noise	P, 2	2,500 GFP-tagged strains	rich / minimal	$5.0 \times 10^3 \ddagger$	1.0×10^4	S288C+tag	–	Measures the distribution rather than the mean, per protein, in two media. Noise is the quantity a per-cell readout recovers and a bulk one destroys, so this sets what a single-cell design should expect to see.
15	Mukherjee 2021 (CRISPRi essential genes x acetic acid) Tolerance / robustness; fitness under acid stress	P, 2	1,100 essential genes	acetic acid vs control	$2.2 \times 10^3 \ddagger$	2.2×10^3	S288C+guide	–	Acetic acid is the dominant hydrolysate stressor, and this reaches the essential genes a deletion collection structurally cannot.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
16	Lian 2017 (CRISPR-AID, beta-carotene) CRISPR library screen; beta-carotene titer	P, 2	combinatorial a/i/d triples	1	200‡	200	engineered-chassis	–	The tri-functional chassis MAGIC was built on, read out on beta-carotene. Small, but it is a combinatorial genotype with a measured product titer, which is the exact record type inverse strain design needs.
17	Hughes 2000 (compendium of expression profiles) Expression / single cell; mRNA log ratio (two-color array)	P, 3	300 mutations and chemical treatments	1	300	1.8×10^6	S288C-KO	–	The original deletion expression compendium, and the oldest independent measurement of the phenotype the built Kemmeren 2014 measures. Ingesting it makes cross-laboratory replication of the deletion transcriptome testable on gene-by-gene profiles fourteen years apart and on different array platforms, which is a check the built set cannot run against itself. The 300 profiles mix deletions, titratable alleles and compound treatments, so the compound arm carries a reference genotype and the mutant arm an S288C-KO one; the split was not confirmed this pass.
18	Lenstra 2011 (chromatin regulator deletion expression) • Expression / single cell; mRNA abundance	P, 3	165 chromatin-regulator deletions	1	165	9.9×10^5	S288C-KO	–	Transcriptomes for the chromatin machinery specifically, from the group behind the built Kemmeren compendium and in the same format. Chromatin regulators are the class whose deletion most reshapes the expression landscape.
19	Sun 2013 (mRNA synthesis and decay rates across deletion strains) Expression / single cell; mRNA level, synthesis rate and decay rate	P, 3	46 deletion strains of mRNA degradation and metabolism genes	1	46	8.3×10^5	S288C-KO	rates derived from metabolic labeling; the rates are stored, not the series	Comparative dynamic transcriptome analysis decomposes a transcript level into the synthesis rate and the decay rate that produce it, across 46 single deletions. That is the transcript-side counterpart of Martin-Perez 2017’s protein half-lives, so the two together give both turnover terms. Its finding also warns what a steady-state compendium hides: a change in degradation rate is generally compensated by a change in synthesis rate, so mRNA level is buffered and two strains with the same level can have different kinetics. Overlap with the other deletion-strain expression sets is on the strain axis, not the measurement: the built Kemmeren 2014 (1,484 deletions), Hughes 2000, Hu 2007 and Lenstra 2011 all store a steady-state level per gene, and none stores a rate, so this row is a second layer on shared strains rather than a second copy of one. The 46 strains are single deletions of mRNA degradation and metabolism genes and are expected to be inside the Kemmeren set by gene identity; the exact shared-strain count needs the GEO strain list, which was not fetched, and the abstract does not say the strains came from that collection.
20	Jariani 2020 (yeast scRNA-seq through lag phase) • Expression / single cell; single-cell transcriptome	P, 3	isogenic wild type	glucose to maltose shift, time course through lag phase	6‡	3.6×10^4	reference-only	time course through the lag phase of a glucose-to-maltose shift	A yeast-adapted 10x protocol that resolves the transcriptional states of individual cells during a carbon-source shift, so the readout is the distribution across a population that is not synchronized rather than its mean. One genotype, so the value is the protocol and the per-cell variance structure a campaign has to budget against.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
21	Guo 2018 (CRISPR-Cas9 tiling of essential genes) • CRISPR library screen; per-guide fitness	P, 3	3e4 tiling guides	1	3.0×10^4 ‡	3.0×10^4	S288C+guide	–	Tiles guides across genes rather than sampling a few per gene, so the readout resolves within-gene functional regions. A finer input axis than a gene-level library.
22	Urbonaite 2021 (yeastDrop-Seq under drug treatment) • Expression / single cell; single-cell transcriptome	P, 3	isogenic wild type	mycophenolic acid, guanine, both, untreated	4	2.4×10^4	reference-only	–	A yeast-optimized Drop-seq with a small chemical condition axis, including a two-compound arm. One genotype, so it enters as a platform reference: the pair of single-drug arms against their combination is a per-cell readout of a two-factor environment, which is the environmental analog of the combinatorial genetic perturbation a Perturb-seq needs.
23	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq) • Expression / single cell; single-cell transcriptome	P, 3	clonal wild type	baseline + stress	2‡	1.2×10^4	reference-only	–	Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.
24	Gasch 2017 (single-cell stress heterogeneity) • Expression / single cell; single-cell transcriptome	P, 3	wild type	stress vs unstressed	2‡	1.2×10^4	reference-only	–	Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.
25	Jaffe 2019 (multiplexed CRISPR interference epistasis) • CRISPR library screen; fitness / genetic interaction	P, 3	1e4 guide pairs	1	1.0×10^4 ‡	1.0×10^4	S288C+guide	–	Two guides per cell, so the perturbation space is combinatorial rather than one edit at a time. This is the shape a Perturb-seq input axis should have, measured here against fitness only.
26	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics) • Expression / single cell; single-cell transcriptome (173,361 cells)	P, 3	wild type	one time course, sampled continuously	1	6.0×10^3	reference-only	one time course, sampled continuously	The second scYeast pre-training set: 173,361 cells from one wild-type strain sampled continuously through a time course without metabolic labeling, so the strain-by-condition count is one and the value is the per-cell distribution of transcriptional states, not a genotype axis. A reference backbone for any single-cell expression head, and the companion of Jackson 2020 on the same platform.
27	Brettner 2024 (ultra-high-throughput yeast scRNA-seq) • Expression / single cell; single-cell transcriptome	P, 3	method	method	–‡	–	reference-only	–	The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.
28	Mulleder 2012 (prototrophic deletion collection) Modality / backbone; strain resource (no phenotype)	P, 3	5,185 prototrophic + 839 titratable essential	1	–‡	–	S288C-KO	–	The background the already-built Mulleder 2016 and Messner 2023 were measured in, and mulleder2016.py already cites it. Listed not as an ingestion target but because that loader records the prototrophy-restoring markers as unmodeled: they are a GeneAddition this schema does not yet carry, and auxotrophy markers distort metabolite pools.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
29	Su 2023 (single-cell transcriptomes under four stresses) Expression / single cell; single-cell transcriptome (117 cells, full length)	P, 4	wild type	isosmotic, hypoosmotic, glucose starvation, amino acid starvation	4	2.4×10^4	reference-only	–	Full-length single-cell RNA-seq of 117 cells across four stress conditions in one strain; scYeast’s stress-state classification benchmark (Fan 2027, Sec. 4.1.6). Tiny, but it is a full-length protocol rather than 3’ tag counting, which matters for isoform-level heads.
30	Wang 2022 (single-cell transcriptomes across replicative aging) Expression / single cell; single-cell transcriptome (125 cells)	P, 4	wild type	2 h, 16 h, 36 h of aging	3	6.6×10^3	reference-only	three ages: 2 h, 16 h and 36 h	Three age groups of one strain, 37, 43 and 45 cells each, about 2,200 genes detected per cell. Small, but it is the only yeast single-cell set with age as the condition axis, and scYeast used it to infer age-specific regulatory relationships (Fan 2027, Sec. 4.1.4).
31	Albert 2018 (eQTL in 1,012 segregants) • Natural variation; mRNA abundance (RNA-seq)	M, 1	1,012 sequenced segregants	1	1.0×10^3	5.8×10^6	segregant-WGS	–	Transcriptomes on a thousand sequenced recombinants, so genotype maps to a genome-wide expression vector rather than one trait. The single best existing analog to a Perturb-seq readout on a high-dimensional genotype axis.
32	Jakobson 2025 (genome-to-proteome map) Natural variation; protein abundance, >6,400 pQTL	M, 1	800 F6 segregants	1	800	1.6×10^6	segregant-WGS	–	Natural alleles mapped to enzyme abundance. A lower-risk engineering lever than a heterologous construct, and the proteome counterpart to the metabolite-QTL panels.
33	Muenzner 2024 (natural-isolate proteome) Natural variation; protein abundance (DIA-MS)	M, 1	796 natural isolates	1	796	1.6×10^6	isolate-WGS	–	Proteomes on the sequenced isolate panel whose transcriptomes Caudal 2024 already supplies. That overlap is the paired anchor the RNA-to-protein inference thesis needs.
34	Aulakh 2025 (genome-scale ionome) Metabolite / precursor; intracellular metal-ion content (ICP-MS)	M, 1	4,800 YKO	1	4.8×10^3	4.8×10^4	S288C-KO	–	The only gene-indexed ionome. Metal-cofactor supply gates iron-sulfur and zinc-dependent pathway enzymes, and no built dataset measures it.
35	Teyssonniere 2024 (species-wide proteome against transcriptome) Natural variation; protein abundance, paired with transcript abundance	M, 2	942 natural isolates	1	942	1.9×10^6	isolate-WGS	–	Quantitative proteomes for 942 sequenced isolates, the same panel whose transcriptomes the built Caudal 2024 supplies, so the strain-level pairing is one to one rather than an overlap. It is also the published answer to what that pairing shows: mRNA and protein correlate weakly at the population gene level, and the variants associated with protein levels overlap those associated with transcript levels by 3 percent. That makes it the calibration for any RNA-to-protein inference rather than only another proteome. DE-DUPLICATION OPEN: it shares authors and a panel with Muenzner 2024 at 796 isolates, and whether the two are independent acquisitions or one re-served is not established here and must be settled before both are built.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
36	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome) Natural variation; transcript, protein and phosphosite abundance	M, 2	112 genomically defined strains (BY x RM panel)	1	112	1.1×10^6	segregant-WGS	–	The only row that measures transcript, protein and protein phosphorylation on one set of sequenced strains, 1,862 proteins and 2,116 phosphopeptides over 988 proteins. The pairing it yields is per strain rather than per gene: for each of the 112 genotypes the mRNA level and the protein level of the same gene are both observed, which is what turns the RNA-to-protein question from a population correlation into a within-strain residual. It also reports that genetic variants changing a protein’s phosphorylation are mostly different from those changing its abundance, so the phosphosite layer is not a proxy for the protein layer.
37	Leutert 2023 (phosphoproteome x 101 conditions) Expression / single cell; phosphosite abundance	M, 2	wild type	101 conditions	101	5.0×10^5	reference-only	–	Post-translational enzyme control acts faster than transcription and is invisible to every other layer here. Condition axis only.
38	Brauer 2008 (growth-rate-controlled chemostat transcriptome) Expression / single cell; mRNA abundance	M, 2	wild type	36 chemostats, 6 limitations	36	2.2×10^5	reference-only	–	Separates growth rate from nutrient identity by design, which no batch-culture dataset can. A large share of any expression response is a growth-rate confound, and this is the reference for removing it.
39	Hackett 2016 (SIMMER multi-omic flux) Expression / single cell; paired metabolome + proteome + flux	M, 2	wild type	25 chemostat states	25	7.5×10^4	reference-only	–	The only jointly measured metabolome, proteome and fluxome. No genotype axis, so it is a mechanism prior rather than training data, but it is the only place the three layers are measured on the same cells.
40	Zhu 2014 (kinase / phosphatase lipidomics) Metabolite / precursor; lipid class and species abundance	M, 2	129 kinase + phosphatase KOs	1	129	2.6×10^4	S288C-KO	–	A gene-indexed lipidome on a regulatory gene set. Complements the already-built da Silveira 2014 lipidomics, which has no kinase axis, and reads out the malonyl-CoA sink directly.
41	Gerke 2017 (urea-cycle mQTL) Natural variation; untargeted metabolite abundance	M, 2	147 diploid segregants	1	147	1.5×10^4	segregant-WGS	–	A second mQTL panel from a different cross (oak x wine), so cross-background generalization of metabolite QTL can be tested rather than assumed.
42	Ambroset 2014 (metabolite QTL) Natural variation; 74 metabolite concentrations (LC-MS/MS)	M, 2	100 segregants	1	100	7.4×10^3	segregant-WGS	–	The founding metabolite-QTL panel. Pairs with Cooper and Mulleder as the natural-variation view of a phenotype space the deletion collection covers by engineering.
43	Blank 2005 (13C metabolic flux) Metabolite / precursor; intracellular flux distribution (13C-MFA)	M, 2	200 viable null mutants	glucose minimal	200†	6.0×10^3	S288C-KO	–	The only real flux measurement tied to single-gene deletions. Everything else in the built set is a concentration or a titer, which is a proxy for flux rather than flux.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
44	Skelly 2013 (expression variation across isolates) Natural variation; transcriptome + proteome	M, 3	22 sequenced isolates	1	22	1.8×10^5	isolate-WGS	–	Transcriptome and proteome on the same isolates, which is the paired anchor the RNA-to-protein inference argument needs. Few strains, but the pairing is measured rather than joined across studies.
45	Airoidi 2016 (nitrogen-limited steady-state and dynamic transcriptome) Expression / single cell; bulk transcriptome	M, 3	wild type, prototrophic	nitrogen sources x steady-state growth rates, plus an upshift time course	20†	1.2×10^5	reference-only	an upshift time course beside the steady states	Chemostat transcriptomes under controlled nitrogen limitation from the same laboratory, and on the same media axis, as Jackson 2020’s NLIM-GLN, NLIM-PRO, NLIM-NH4 and NLIM-UREA conditions. That shared axis is what makes the wild-type baseline of the single-cell atlas separable from the deletion effect, and it is the nitrogen counterpart of Brauer 2008’s carbon-limited growth-rate series. Counts are from the citation rather than the released series and need confirming.
46	McManus 2014 (ribosome profiling, allele-specific) Modality / backbone; ribosome occupancy paired with mRNA abundance	M, 3	S. cerevisiae, S. paradoxus and their F1 hybrid	1	3	3.3×10^4	reference-only	–	Translation efficiency measured gene by gene alongside the mRNA abundance of the same gene, which is the first of the two terms that make protein level differ from transcript level. Its result is the reason the term is needed: ribosome occupancy is more conserved than transcript abundance, so translation buffers divergence in mRNA rather than tracking it. The S. paradoxus arm is out of species and would be stored separately or dropped; the cerevisiae arm and the hybrid’s cerevisiae alleles are in scope.
47	Martin-Perez 2017 (protein half-lives) Modality / backbone; protein turnover rate	M, 3	wild type, exponential growth	1	1	3.2×10^3	reference-only	half-lives derived from a labeling time course; the rates are stored	Turnover rates for 3,160 proteins, the second term that makes protein level differ from transcript level. Half-lives span three orders of magnitude and are not normally distributed, so a single global scaling from mRNA to protein cannot be right, and this is the per-gene coefficient that replaces it. The paper also reports that localization, complex membership and connectivity predict turnover better than sequence features do, which names the covariates such a model needs.
48	Boer 2010 (metabolome across nutrient limitations) Metabolite / precursor; intracellular metabolite concentrations	M, 3	wild type	nutrient limitations x growth rates	25	2.5×10^3	reference-only	–	The metabolite counterpart to Brauer on the same chemostat design, so metabolite pools are separated from growth rate the same way. Covers central-carbon nodes the deletion metabolomics rows measure at low power.
49	Tengolics 2024 (domestication metabolome) Natural variation; 19 amino acids + 78 other metabolites	M, 3	17 S. cerevisiae populations	1	17	1.6×10^3	isolate-WGS	–	Shows which metabolic traits already moved under domestication, which is a prior on which are tractable to engineer further. Small n.
50	Eder 2020 (flux QTL) Natural variation; modeled intracellular flux	M, 3	segregant panel (n unconfirmed)	wine fermentation	100†	100	segregant-WGS	–	Flux as a QTL phenotype, with PDB1 and VID30 validated. The label is modeled rather than measured, which caps how far it can be trusted.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
Ten extra, in rank order: rows 51–60 replace a row above that proves unreachable.									
51	Foss 2007 (BY x RM segregant proteome) Natural variation; protein abundance (label-free MS)	M, 3	BY x RM segregants; the count is not stated in the abstract and was not confirmed	1	– [‡]	–	segregant-WGS	–	The first segregant proteome, and the independent replicate of Grossbach 2022 on the same cross fifteen years earlier and on a different platform. Its finding is the one this whole group turns on: the loci that influence protein abundance differ from those that influence transcript levels. Ranked below Grossbach because Grossbach measures the transcript layer on the same strains and this does not, and because neither the segregant count nor the proteome depth was confirmed this pass.
52	Yu 2021 (proteome and metabolome under nitrogen limitation) Metabolite / precursor; proteome + metabolome	M, 4	wild type	nitrogen-limited chemostats	10 [‡]	2.5 × 10 ⁴	reference-only	–	Nitrogen limitation is the standard trigger for redirecting carbon to product, and this measures both layers under it. Condition axis only, and small.
53	de Boer 2020 (100M random promoters) • Regulatory DNA; expression driven (YFP sort-seq)	S, 1	>100M random promoters	complex + defined	1.0 × 10 ⁸	1.0 × 10 ⁸	S288C+ reporter-locus	–	The largest sequence-to-phenotype dataset in yeast by orders of magnitude, and every genotype is an exactly known 80mer. The perturbation space is continuous rather than a gene list, which is what a model of regulatory grammar needs and no deletion collection can supply.
54	Vaishnav 2022 (regulatory DNA fitness landscape) • Regulatory DNA; expression + inferred fitness	S, 1	2e7 promoter variants	1	2.0 × 10 ⁷ [‡]	2.0 × 10 ⁷	S288C+ reporter-locus	–	Extends de Boer from measurement to a fitness landscape, and tests native and natural-isolate promoter variants against the random library. The bridge between synthetic sequence space and the natural variation the isolate panels carry.
55	Lee 2014 (HIP-HOP fitness signatures)^B Tolerance / robustness; chemogenomic fitness score	S, 1	11,000 (het + hom)	3,356 compounds	1.3 × 10 ⁷	1.3 × 10 ⁷	S288C-KO	–	Largest chemical-genetic matrix in existence, on the exact YKO genotype axis Costanzo and Kuzmin already use. Biggest single instance-count gain available. BLOCKED: WS15 already attempted it and it stands as awaiting author matrices, so the Science SI and the lab portal did not yield per-strain values. Unblocking is an author request, not a loader.
56	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants) • Natural variation; fitness (Bar-seq)	S, 1	100,000 barcoded segregants	8 conditions	8.0 × 10 ⁵ [‡]	8.0 × 10 ⁵	segregant-WGS	–	The largest genotyped recombinant panel in yeast, each segregant barcoded and low-coverage sequenced against known parents. Two orders of magnitude more genotypes than Bloom 2013 on the same kind of cross, which is what makes small-effect and epistatic loci detectable.
57	Galardini 2019 (four backgrounds x 38 conditions) Natural variation; colony-size fitness (S-score)	S, 1	3,786 KOs x 4 backgrounds	38 conditions	5.8 × 10 ⁵ [‡]	5.8 × 10 ⁵	isolate-WGS	–	The same knockout built independently in four sequenced backgrounds. Quantifies the 18.5 percent of deletion phenotypes that do not transfer, which is the generalization risk the whole substrate is exposed to.

Table 1, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
58	Parsons 2006 (bioactive-compound profiling) Tolerance / robustness; hypersensitivity score	S, 1	5,000 viable YKO	82 compounds	$4.1 \times 10^5 \dagger$	4.1×10^5	S288C-KO	–	Full-collection strain coverage against a modest compound panel, the complement to the diagnostic-panel screens above.
59	Dutta 2026 (barcoded natural-isolate chemical response) Natural variation; pooled-barcode fitness	S, 1	520 natural isolates	>600 compounds	$3.1 \times 10^5 \dagger$	3.1×10^5	isolate-WGS	–	Natural sequence diversity crossed with a chemogenomic panel, on isolates drawn from the sequenced 1,011 collection. The same strains already carry Caudal 2024 transcriptomes and Muenzner 2024 proteomes, so it is a three-modality join on one genotype axis.
60	Li 2016 (tRNA fitness landscape) • Deep mutational scan; fitness	S, 1	65,000 tRNA variants	1	$6.5 \times 10^4 \ddagger$	6.5×10^4	S288C+ designed-edit	–	A near-complete fitness landscape of one native yeast gene. Every genotype is a known sequence and the sampling is exhaustive rather than one allele per gene, which is the opposite of what a deletion collection provides.

Table 2: Sources for Table 1, in the same order. *Why it is ordered here* is the band reason for a perturb-seq or molecular-layers row, and the tier and measurement rule for a scale row; what the row buys is the *Why* column of Table 1. *Data* is where the per-record values live; entries marked unconfirmed were not fetched live and must be checked before a loader is written. Every link is clickable.

# Dataset, citation and link	Why it is ordered here	Data
1 Boocock 2025 (single-cell eQTL mapping) Boocock J, Alexander N, Alamo Tapia L, Walter-McNeill L, Patel SP, Munugala C, Bloom JS, Kruglyak L. eLife 2025;13:e95566. doi:10.7554/eLife.95566	Genotype recovered per cell from the cell's own reads, crossed with a per-cell transcriptome. The largest existing yeast dataset in the quadrant the campaign targets.	mirrored as boocockSinglecellEQTLMapping2025; GEO (unconfirmed)
2 Hale 2024 (CRISPRi x natural variation) Hale JJ, Matsui T, Goldstein I, et al., Kruglyak L. Nat Commun 2024;15:4234. doi:10.1038/s41467-024-48626-1	The one released yeast library that crosses a CRISPR interference guide set with a sequenced genetic background, which is the input axis a campaign has to plan against.	SRA PRJNA986287 + Figshare
3 Jackson 2020 (TF-deletion single-cell atlas) Jackson CA, Castro DM, Saldi GA, Bonneau R, Gresham D. eLife 2020;9:e51254. doi:10.7554/eLife.51254	Transcribed genotype barcodes with a per-cell transcriptome readout, fully crossed over 11 conditions. The design a yeast Perturb-seq is specified against.	GEO GSE125162; also eLife Source code 2 (103118_SS_Data.tsv.gz). Confirmed from the mirrored paper. The released matrix carries 38,225 cells; the abstract says 38,285 and the discussion 38,255, so the deposit governs. The authoritative 72-strain genotype list is Supplementary file 1 Table S2, an Excel file not held in the mirror.
4 Puddu 2019 (WGS of the deletion collection) Puddu F, et al., Jackson SP. Nature 2019;573:416-420. doi:10.1038/s41586-019-1549-9	Whole-genome sequence for every strain in the deletion collection. A pooled campaign reads a barcode and infers a genotype; this is the measurement of how often that inference is wrong, and Jackson 2020 built its own strains rather than use the collection for related reasons.	ENA/SRA (accession unconfirmed)
5 N'Guessan 2025 (segregant scRNA-seq eQTL) N'Guessan A, Boocock J, Kruglyak L, Albert FW. eLife 2025. PMC12303567. PMC12303567	About 4,500 sequenced segregants profiled by single-cell RNA-seq, so genotype and transcriptome are measured in the same cell.	eLife data availability; segregant panel from Bloom lineage
6 Hackett 2020 (IDEA inducible-TF transcriptome time series) Hackett SR, Baltz EA, Coram M, Wranik BJ, Kim G, Baker A, Fan M, Hendrickson DG, Berndt M, McIsaac RS. Mol Syst Biol 2020;16:e9174. doi:10.15252/msb.20199174	Induction rather than deletion, with a transcriptome followed over time. Jackson 2020 names transient induction as the perturbation modality most likely to produce a detectable expression response.	Published with the paper (Mol Syst Biol SI); accession unconfirmed from the mirror
7 Hu 2007 (TF deletion expression compendium) Hu Z, Killion PJ, Iyer VR. Nat Genet 2007;39:683-687. doi:10.1038/ng2012	A designed single-gene perturbation crossed with a transcriptome over 269 transcription-factor deletions, which is the bulk form of what the campaign measures per cell, and the widest regulator coverage available.	GEO (unconfirmed)
8 Dong 2021 (MAGIC + SAM biosensor) Dong C, Schultz JC, Liu W, Lian J, Huang L, Xu Z, Zhao H. Metab Eng 2021;66:319-327. doi:10.1016/j.ymben.2021.03.005	A genome-wide CRISPR activation, interference and deletion library sorted on a metabolite biosensor, so the label is product concentration rather than fitness. The pattern a sorted Perturb-seq would reuse.	Metab Eng SI (accession unconfirmed)
9 Momen-Roknabadi 2020 (inducible CRISPRi library) Momen-Roknabadi A, Oikonomou P, Zegans M, Tavazoie S. Commun Biol 2020;3:723. doi:10.1038/s42003-020-01452-9	A genome-scale inducible CRISPR interference library with a released per-guide matrix. Induction control is what lets a knockdown be applied after a cell is captured rather than during outgrowth.	Addgene + Commun Biol SI

Table 2, continued

#	Dataset, citation and link	Why it is ordered here	Data
10	McGlinchy 2021 (genome-scale CRISPRi library) McGlinchy NJ, Meacham ZA, Reynaud KK, Muller R, Baum R, Ingolia NT. BMC Genomics 2021;22:205. doi:10.1186/s12864-021-07518-0	The second independently designed genome-scale yeast CRISPR interference library with a released per-guide matrix, so guide design and gene effect can be separated.	ingolia-lab.org/yeast-crispri + Addgene + BMC SI
11	Bao 2018 (CHAnGE single-nucleotide library) Bao Z, Hamedirad M, Xue P, Xiao H, Tasan I, Chao R, Liang J, Zhao H. Nat Biotechnol 2018;36:505-508. doi:10.1038/nbt.4132	A guide-indexed genome-wide library whose edits sit below the open reading frame, so the perturbation is an allele rather than a null.	Nat Biotechnol SI (raw sequencing location unconfirmed)
12	Crook 2016 (tunable RNAi, isobutanol + 1-butanol) Crook N, Sun J, Morse N, Schmitz A, Alper HS. Appl Microbiol Biotechnol 2016;100:10005-10018. PMID 27654654. PMID 27654654	A dose-graded knockdown axis. Graded rather than binary perturbation is what turns a per-cell readout into a dose-response curve.	AMB SI (DOI is an open verification item)
13	Roy 2018 (multiplexed precision editing) Roy KR, Smith JD, Vonesch SC, et al., St Onge RP. Nat Biotechnol 2018;36:512-520. doi:10.1038/nbt.4137	Multiplexed designed edits per cell, which is combinatorial input in the sense the Perturb-seq specification requires.	Nat Biotechnol SI (accession unconfirmed)
14	Newman 2006 (protein noise, GFP library) Newman JRS, Ghaemmaghami S, Ihmels J, Breslow DK, Noble M, DeRisi JL, Weissman JS. Nature 2006;441:840-846. doi:10.1038/nature04785	Per-cell protein abundance distributions across a genome-scale tagged collection. Expression noise per gene is what sets how large a perturbation effect has to be before a per-cell readout can resolve it.	Nature SI
15	Mukherjee 2021 (CRISPRi essential genes x acetic acid) Mukherjee V, Lind U, St Onge RP, Blomberg A, Nystrom T. mSystems 2021;6:e00410-21. doi:10.1128/mSystems.00410-21	Knockdown reaches the essential genes a deletion collection cannot hold, which is a third of the genome a deletion-based campaign would miss.	mSystems open-access SI
16	Lian 2017 (CRISPR-AID, beta-carotene) Lian J, Hamedirad M, Hu S, Zhao H. Nat Commun 2017;8:1688. doi:10.1038/s41467-017-01695-x	Three perturbation modalities in one cell, activation, interference and deletion, which is combinatorial input by modality rather than by gene.	Nat Commun SI
17	Hughes 2000 (compendium of expression profiles) Hughes TR, Marton MJ, Jones AR, Roberts CJ, Stoughton R, et al., Friend SH. Cell 2000;102:109-126. doi:10.1016/S0092-8674(00)00015-5	The original designed-perturbation expression compendium, and the independent measurement that makes cross-laboratory replication of the deletion transcriptome testable against the built Kemmeren 2014.	Rosetta compendium, distributed with the paper; not fetched
18	Lenstra 2011 (chromatin regulator deletion expression) Lenstra TL, Benschop JJ, Kim T, et al., Holstege FCP. Mol Cell 2011;42:536-549. doi:10.1016/j.molcel.2011.03.026	The same design over chromatin regulators rather than sequence-specific factors, so the two together cover both halves of transcriptional control by deletion.	GEO (unconfirmed)
19	Sun 2013 (mRNA synthesis and decay rates across deletion strains) Sun M, Schwalb B, Pirkel N, Maier KC, Schenk A, Failmezger H, Tresch A, Cramer P. Mol Cell 2013;52:52-62. doi:10.1016/j.molcel.2013.09.010	A designed single-gene perturbation crossed with a transcriptome, and the only one that resolves the level into a synthesis rate and a decay rate. Rates are what a per-cell snapshot cannot recover on its own.	GEO, series not fetched this pass

Table 2, continued

#	Dataset, citation and link	Why it is ordered here	Data
20	Jariani 2020 (yeast scRNA-seq through lag phase) Jariani A, Vermeersch L, Cerulus B, Perez-Samper G, Voordeckers K, Van Brussel T, Thienpont B, Lambrechts D, Verstrepen KJ. eLife 2020;9:e55320. doi:10.7554/eLife.55320	Per-cell states through a carbon-source shift, so the readout is the distribution across an unsynchronized population rather than its mean.	GEO GSE144820 and GSE116246; from the collection sweep, not fetched
21	Guo 2018 (CRISPR-Cas9 tiling of essential genes) Guo X, Chavez A, Tung A, et al., Church GM, Shalem O. Nat Biotechnol 2018;36:540-546. doi:10.1038/nbt.4147	Guide tiling produces a graded allelic series across one gene, so guide position rather than gene identity carries the effect.	Nat Biotechnol SI (unconfirmed)
22	Urbonaite 2021 (yeastDrop-Seq under drug treatment) Urbonaite G, Lee JTH, Liu P, Parras GG, Hemberg M, Acar M. Commun Biol 2021;4:1372. doi:10.1038/s42003-021-02895-4	Single-drug arms against their combination with a per-cell readout, the environmental analog of a combinatorial genetic perturbation.	GEO GSE165686; Zenodo 10.5281/zenodo.4767298 and 10.5281/zenodo.4762526; from the collection sweep, not fetched
23	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq) Nadal-Ribelles M, Islam S, Wei W, et al., Posas F, Steinmetz LM. Nat Microbiol 2019;4:683-692. doi:10.1038/s41564-018-0346-9	The high-sensitivity yeast protocol, strand and isoform aware. Sensitivity per cell is the constraint on how small a perturbation effect a campaign can resolve.	Nat Microbiol SI /GEO
24	Gasch 2017 (single-cell stress heterogeneity) Gasch AP, Yu FB, Hose J, et al., Quake SR. PLoS Biol 2017;15:e2004050. doi:10.1371/journal.pbio.2004050	Separates intrinsic from extrinsic per-cell variation in an isogenic population under stress, which is the null a perturbation effect is measured against.	PLoS Biology SI /GEO
25	Jaffe 2019 (multiplexed CRISPR interference epistasis) Jaffe M, Sherlock G, Levy SF. G3 2019;9:2843-2852. doi:10.1534/g3.119.400323	Two knockdowns in the same cell with a measured interaction, which is the only combinatorial CRISPR interference design located in yeast.	G3 open-access SI (unconfirmed)
26	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics) Jackson CA, Beheler-Amass M, Tjarnberg A, Suresh I, Hickey AS, Bonneau R, Gresham D. bioRxiv 2023. 10.1101/2023.09.21.558277. doi:10.1101/2023.09.21.558277	The platform companion of Jackson 2020: 173,361 cells of one strain, which is what sets the per-cell variance a perturbation has to exceed.	GEO GSE242556 (from the scYeast Methods, Sec. 4.1.3)
27	Brettner 2024 (ultra-high-throughput yeast scRNA-seq) Brettner L, et al. Yeast 2024. doi:10.1002/yea.3927	Combinatorial barcoding in yeast at 96 to 384 multiplexed genotypes or environments per run. The throughput route by which a campaign becomes affordable.	Yeast (Wiley) SI
28	Mulleder 2012 (prototrophic deletion collection) Mulleder M, Capuano F, Pir P, et al., Ralser M. Nat Biotechnol 2012;30:1176-1178. doi:10.1038/nbt.2442	The prototrophic deletion collection. Jackson 2020 used a prototrophic background because auxotrophy blocks minimal and nitrogen-limited media, so this is the strain resource a campaign in defined media needs.	EUROSCARF; Addgene kit 40276
29	Su 2023 (single-cell transcriptomes under four stresses) Su Y, Xu C, Shea J, DeStephanis D, Su Z. BMC Genomics 2023;24:88. doi:10.1186/s12864-023-09184-w	Full-length rather than three-prime tag counting, so isoform-level readout is on the table for a campaign that needs it.	GEO GSE201386 (from the scYeast Methods)

Table 2, continued

#	Dataset, citation and link	Why it is ordered here	Data
30	Wang 2022 (single-cell transcriptomes across replicative aging) Wang J, Sang Y, Jin S, Wang X, Azad GK, McCormick MA, Kennedy BK, Li Q, Wang J, Zhang X, et al. Aging Cell 2022;21:e13712. doi:10.1111/acer.13712	The only yeast single-cell set with age as the condition axis, which is a covariate any pooled campaign carries whether or not it measures it.	GEO GSE210032 (from the scYeast Methods)
31	Albert 2018 (eQTL in 1,012 segregants) Albert FW, Bloom JS, Siegel J, Day L, Kruglyak L. eLife 2018;7:e35471. doi:10.7554/eLife.35471	The transcriptome half of the segregant panel that Jakobson 2025, Gerke 2017 and Eder 2020 measure protein, metabolite and flux on.	GEO /eLife data availability (unconfirmed)
32	Jakobson 2025 (genome-to-proteome map) Jakobson CM, et al. Science 2025;390:eadu3198. doi:10.1126/science.adu3198	Protein abundance across the same sequenced segregant panel Albert 2018 profiles by transcriptome, so protein and transcript quantitative trait loci are measurable on one genotype set.	Science data-availability statement (accession unconfirmed)
33	Muenzner 2024 (natural-isolate proteome) Muenzner J, et al., Ralser M. Nature 2024;630:149-157. doi:10.1038/s41586-024-07442-9	796 proteomes drawn from the sequenced 1,011-isolate panel that the supported Caudal 2024 transcriptomes also come from.	PRIDE PXD048219
34	Aulakh 2025 (genome-scale ionome) Aulakh SK, et al. Cell Syst 2025;16:101319. doi:10.1016/j.cels.2025.101319	A metabolic readout on the whole deletion collection, which is the genotype axis the supported expression compendium already covers.	Cell Systems SI (accession unconfirmed)
35	Teyssonniere 2024 (species-wide proteome against transcriptome) Teyssonniere EM, Trebulle P, Muenzner J, Loegler V, Ludwig D, Amari F, Mulleder M, Friedrich A, Hou J, Ralser M, Schacherer J. Proc Natl Acad Sci USA 2024;121:e2319211121. doi:10.1073/pnas.2319211121	Proteomes for 942 isolates, the panel whose transcriptomes the built Caudal 2024 supplies, so the strain-level pairing is one to one.	PNAS SI; PMC11087752. PRIDE identifier not fetched this pass
36	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome) Grossbach J, Gillet L, Clement-Ziza M, Schmalohr CL, Schubert OT, Schutter M, Mawer JSP, Barnes CA, Bludau I, Weith M, Tessarz P, Graef M, Aebersold R, Beyer A. Mol Syst Biol 2022;18:e10712. doi:10.15252/msb.202110712	Three layers on one set of 112 sequenced strains, so the transcript and the protein of a gene are observed in the same genotype and the RNA to protein question becomes a within-strain residual.	Mol Syst Biol SI + PRIDE, identifier not fetched this pass
37	Leutert 2023 (phosphoproteome x 101 conditions) Leutert M, et al., Villen J. Nat Struct Mol Biol 2023;30:1761-1773. doi:10.1038/s41594-023-01115-3	Enzyme regulation acts faster than transcription, so a phosphosite layer over 101 conditions is what explains flux changes an expression table cannot.	PRIDE PXD034997
38	Brauer 2008 (growth-rate-controlled chemostat transcriptome) Brauer MJ, Huttenhower C, Airoidi EM, et al., Botstein D. Mol Biol Cell 2008;19:352-367. doi:10.1091/mbc.e07-08-0779	Expression at controlled growth rate under each nutrient limitation, which is the condition axis Boer 2010 and Hackett 2016 measure metabolites and flux on.	GEO (unconfirmed)

Table 2, continued

#	Dataset, citation and link	Why it is ordered here	Data
39	Hackett 2016 (SIMMER multi-omic flux) Hackett SR, Zanutelli VRT, Xu W, et al., Rabinowitz JD. Science 2016;354:aaf2786. doi:10.1126/science.aaf2786	Flux, metabolite and transcript measured in one study under the same nutrient limitations, so the join is internal rather than across papers.	Science SI (accession unconfirmed)
40	Zhu 2014 (kinase / phosphatase lipidomics) Zhu Z, Loewen CJR. Mol Biol Cell 2014. PMC4196872. PMC4196872	A lipidome over 129 signaling deletions, most of which carry an expression profile in the supported compendium.	Mol Biol Cell SI Table S4 (Excel)
41	Gerke 2017 (urea-cycle mQTL) Gerke J, et al., Fay JC. Genetics 2017;206:2199. doi:10.1534/genetics.117.201107	Metabolite quantitative trait loci on a segregant cross that also has an expression map, so a metabolite locus can be read through its transcript.	Genetics (GSA) SI
42	Ambroset 2014 (metabolite QTL) Ambroset C, et al., Fay JC. PLoS Genet 2014;10:e1004142. doi:10.1371/journal.pgen.1004142	A 74-metabolite panel on a sequenced segregant panel, the widest metabolite vector available on a recombinant genotype axis.	PLOS Genetics SI
43	Blank 2005 (13C metabolic flux) Blank LM, Kuepfer L, Sauer U. Genome Biol 2005;6:R49. doi:10.1186/gb-2005-6-6-r49	The only row measuring flux rather than a concentration, on deletion mutants that also have a transcriptome in the supported set.	Genome Biology open-access SI
44	Skelly 2013 (expression variation across isolates) Skelly DA, Merrihew GE, Riffle M, et al., MacCoss MJ, Akey JM. Genome Res 2013;23:1496-1504. doi:10.1101/gr.155762.113	Transcriptome and proteome on the same 22 strains, which is the only row where the cheap and expensive layers are paired within one experiment rather than joined across two.	Genome Research SI (unconfirmed)
45	Airoidi 2016 (nitrogen-limited steady-state and dynamic transcriptome) Airoidi EM, Miller D, Athanasiadou R, Brandt N, Abdul-Rahman F, Neymotin B, Hashimoto T, Bahmani T, Gresham D. Mol Biol Cell 2016;27:1383-1396. doi:10.1091/mbc.E14-05-1013	Expression under the exact nitrogen-limited conditions that Jackson 2020 profiles single cells in, so the deletion effect separates from the medium effect.	GEO, series not confirmed this pass
46	McManus 2014 (ribosome profiling, allele-specific) McManus CJ, May GE, Spealman P, Shteyman A. Genome Res 2014;24:422-430. doi:10.1101/gr.164996.113	Translation efficiency per gene alongside the mRNA abundance of that gene, which is the first of the two terms that separate protein level from transcript level.	GEO, series not fetched this pass
47	Martin-Perez 2017 (protein half-lives) Martin-Perez M, Villen J. Cell Syst 2017;5:283-294.e5. doi:10.1016/j.cels.2017.08.008	Protein turnover per gene, the second of those two terms, spanning three orders of magnitude and therefore not replaceable by a global constant.	Cell Systems SI; PRIDE identifier not fetched this pass
48	Boer 2010 (metabolome across nutrient limitations) Boer VM, Crutchfield CA, Bradley PH, Botstein D, Rabinowitz JD. Mol Biol Cell 2010;21:198-211. doi:10.1091/mbc.e09-07-0597	The metabolite half of the chemostat nutrient-limitation series whose transcriptome half is Brauer 2008, same laboratory and same conditions.	MBoC SI (unconfirmed)
49	Tengolics 2024 (domestication metabolome) Tengolics R, Szappanos B, Mulleder M, et al., Ralser M, Papp B. PNAS 2024;121:e2313354121. doi:10.1073/pnas.2313354121	Metabolite levels across sequenced isolates, the panel the supported Caudal 2024 transcriptomes and Muenzner 2024 proteomes also sit on.	PNAS Dataset S11

Table 2, continued

#	Dataset, citation and link	Why it is ordered here	Data
50	Eder 2020 (flux QTL) Eder M, Nidelet T, Sanchez I, Camarasa C, Legras JL, Dequin S. PMID 32034164. PMID 32034164	Flux mapped to segregant genotypes, which is the flux counterpart of the expression and protein maps on the same panel type.	Journal SI (venue and accession unconfirmed)
51	Foss 2007 (BY x RM segregant proteome) Foss EJ, Radulovic D, Shaffer SA, Ruderfer DM, Bedalov A, Goodlett DR, Kruglyak L. Nat Genet 2007;39:1369-1375. doi:10.1038/ng.2007.22	The independent replicate of Grossbach 2022 on the same cross, and the first measurement that protein-level loci differ from transcript-level loci.	Nat Genet supplementary tables; not fetched
52	Yu 2021 (proteome and metabolome under nitrogen limitation) Yu R, Vorontsov E, Sihlbom C, Nielsen J. eLife 2021;10:e65722. doi:10.7554/eLife.65722	Both layers measured in one study under nitrogen limitation, so it calibrates the protein-to-metabolite step that cross-study joins assume.	eLife data availability (unconfirmed)
53	de Boer 2020 (100M random promoters) de Boer CG, Vaishnav ED, Sadeh R, Abeyta EL, Friedman N, Regev A. Nat Biotechnol 2020;38:56-65. doi:10.1038/s41587-019-0315-8	Scale band: tier 1, ordered by measurements (100,000,000).	GEO (de Boer 2020); DREAM 2022 release of 6,739,258 promoters
54	Vaishnav 2022 (regulatory DNA fitness landscape) Vaishnav ED, de Boer CG, Molinet J, Yassour M, Fan L, Adiconis X, Thompson DA, Levin JZ, Cubillos FA, Regev A. Nature 2022;603:455-463. doi:10.1038/s41586-022-04506-6	Scale band: tier 1, ordered by measurements (20,000,000).	GEO per the Nature data-availability statement (unconfirmed)
55	Lee 2014 (HIP-HOP fitness signatures) Lee AY, St Onge RP, Proctor MJ, et al., Giaever G, Nislow C. Science 2014;344:208-211. doi:10.1126/science.1250217	Scale band: tier 1, ordered by measurements (13,000,000).	Science SI + Nislow/Giaever portal, neither of which yielded per-strain matrices; awaiting author matrices per the WS15 roadmap
56	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants) Nguyen Ba AN, Lawrence KR, Rego-Costa A, Gopalakrishnan S, Temko D, Michor F, Desai MM. eLife 2022;11:e73983. doi:10.7554/eLife.73983	Scale band: tier 1, ordered by measurements (800,000).	eLife data availability; SRA (unconfirmed)
57	Galardini 2019 (four backgrounds x 38 conditions) Galardini M, Busby BP, Vieitez C, Dunham AS, Typas A, Beltrao P. Mol Syst Biol 2019;15:e8831. doi:10.15252/msb.20198831	Scale band: tier 1, ordered by measurements (575,472).	GEO GSE123118 + github.com/mgalardini/2018koyeast
58	Parsons 2006 (bioactive-compound profiling) Parsons AB, Lopez A, Givoni IE, et al., Boone C. Cell 2006;126:611-625. doi:10.1016/j.cell.2006.06.040	Scale band: tier 1, ordered by measurements (410,000).	Cell SI tables (accession unconfirmed)
59	Dutta 2026 (barcoded natural-isolate chemical response) Dutta A, Garin M, Loegler V, Brach G, Friedrich A, Yoshimura M, Hirano H, Osada H, Boone C, Yashiroda Y, Hou J, Schacherer J. Nat Commun 2026. doi:10.1038/s41467-026-73532-z	Scale band: tier 1, ordered by measurements (312,000).	Nat Commun SI; isolates from the 1,011 panel (ENA)

Table 2, continued

# Dataset, citation and link	Why it is ordered here	Data
60 Li 2016 (tRNA fitness landscape) Li C, Qian W, Maclean CJ, Zhang J. Science 2016;352:837-840. doi:10.1126/science.aae0568	Scale band: tier 1, ordered by measurements (65,000).	Science SI (unconfirmed)

3 What each candidate joins to ✗

A dataset’s value in a substrate is not only what it measures. It is also what becomes measurable once it sits beside what is already held. Table 3 names 110 such pairs across the sixty rows of Table 1, every row carrying at least one, each with the axis the two share.

3.1 A join needs a shared axis, not a shared theme ✗

The **join key** column is required rather than decorative. Two metabolomics papers are not joinable because both measure metabolites; they are joinable when they measure overlapping analytes on overlapping strains. The keys in use are the deletion collection, the sequenced 1,011-isolate panel, a segregant panel from one cross, a guide library over one gene set, a media formulation, a compound set, and a single gene. A pair with none of these is a theme and is not listed.

Fifty-eight of the 110 pairs, on 42 of the sixty rows, have a partner that is already built, so that join becomes runnable on the day the candidate is ingested rather than after two ingestions. Those are the cheapest experiments in the table and none of them has been run.

Two kinds of key are distinguished in the table and the difference decides what a join can do. A *strain-level* key means both layers were measured on the same genotypes, so the join is a paired observation and the residual between the layers is per strain. A *gene-level* key means one side is a single genome-wide coefficient, a half-life or a translation rate, that applies to every strain alike. A gene-level layer cannot explain why two strains differ; it explains why two genes differ, and the two are not interchangeable.

3.2 The joins that motivate the ordering ✗

Four groups carry most of the weight, and each is a specific question rather than an accumulation.

Transcript against protein, and the two terms between them. Steady-state transcript and steady-state protein are both well covered, by the built Kemmeren 2014 and Messner 2023 and by several rows here, and they disagree. Four rows make the disagreement something a model can carry a term for rather than absorb as error. Grossbach 2022 measures transcript, protein and phosphosite on one set of 112 sequenced strains, so the residual between transcript and protein is observed within a genotype instead of across a population. Teyssonniere 2024 does the strain-level version at species scale, 942 isolate proteomes against the built Caudal 2024 transcriptomes on the same panel, and reports what the pairing shows: the two correlate weakly and the variants associated with each overlap by 3 percent. The remaining two are gene-level. McManus 2014 gives translation efficiency beside mRNA abundance for 5,474 genes and finds ribosome occupancy more conserved than transcript abundance, so translation buffers rather than tracks. Martin-Perez 2017 gives turnover for 3,160 proteins, spanning three orders of magnitude, which is why a single global mRNA-to-protein scaling cannot be right. Sun 2013 supplies the transcript-side counterpart, decomposing mRNA level into synthesis and decay across 46 deletion strains.

Cross-laboratory replication of the deletion transcriptome. Kemmeren 2014 is built and is the largest genotype-indexed transcriptome with designed perturbations, and nothing in the built set can check it. Hughes 2000, Hu 2007 and Sun 2013 all profile deletion strains by expression, in three other laboratories and across fourteen years of array and sequencing platforms, so agreement can be measured gene by gene on shared strains rather than assumed. That check matters here specifically: the built Nadal-Ribelles 2025 does not replicate its own genotype assignment across batches, so a second opinion on what a deletion does to a transcriptome is not a formality.

Metabolism against expression on one genotype axis. Blank 2005 measures flux through central carbon metabolism on deletion mutants that already carry an expression profile in the supported Kemmeren 2014 compendium, which is the only place in this table where the transcript-to-flux step can be fitted rather than assumed. Cooper 2010 measures the same amino-acid pools as the supported Mulleder 2016 by a different platform on an overlapping genotype set, which separates biology from batch. Aulakh 2025 adds a second metabolic panel, the ionome, on the whole collection. Above them sit Jakobson 2025 and Albert 2018, protein and transcript variation on one sequenced segregant panel, and Muenzner 2024 against the supported Caudal 2024, proteome and transcriptome on one isolate panel.

A condition axis measured on more than one layer. Brauer 2008 and Boer 2010 are the transcript and metabolite halves of one chemostat series, same laboratory and same limitations, and Hackett 2016 adds flux over the same conditions. Airolidi 2016, new to this pass, supplies the nitrogen-limited counterpart, and its media are the media Jackson 2020 profiles single cells in. That last pairing is what makes a deletion effect in the single-cell atlas a contrast against a wild-type baseline in the same medium rather than an absolute.

One perturbation, two modalities. Mukherjee 2021 against the supported Mormino 2022 is acetic acid by knock-down at genome scale against twelve genes. Dong 2021 against the supported Lian 2019 is one tri-functional library read out by growth selection and by a metabolite biosensor, which is how a tolerance screen becomes a production screen. McGlincy 2021 against Momen-Roknabadi 2020 is two independently designed CRISPR interference libraries over the same genes, which is the cheapest decisive test of whether a model has learned gene function or guide design.

The same genotype read out in bulk and per cell. Jackson 2020 against the supported Kemmeren 2014 is the same transcription-factor deletions as a population average and as a per-cell distribution, which is the only available calibration of what pseudobulk loses. Hu 2007 extends the bulk side to 269 regulators. Boocock 2025 against Albert 2018 is the same cross measured in bulk and per cell.

None of these transfers has been measured. Each is a hypothesis about what a model could learn, and each becomes a run once both halves are held.

Table 3: Named joins between a candidate and either a supported dataset or another candidate, for the sixty rows of Table 1. # is the candidate’s rank there. A partner in **bold** is already built, so that row’s join becomes runnable the moment the candidate is ingested; an unbolded partner is itself a candidate, so the join costs two ingestions. *Join key* is the shared axis that makes the pair addressable; a pair with no such axis is a theme and is not listed. None of these transfers has been measured.

#	Candidate	Partner	Join key	What the join yields
1	Boocock 2025 (single-cell eQTL mapping)	Nadal-Ribelles 2025 (Perturb-seq) Albert 2018 (eQTL in 1,012 segregants) Jakobson 2025 (genome-to-proteome map)	single-cell transcriptome readout, one genotype per cell same BY by RM cross, bulk against single cell segregant genotype class	One dataset carries genotypes made by recombination and the other genotypes made by deletion, on the same readout, which is the test of whether a per-cell expression model transfers between the two. The same expression quantitative trait loci measured in bulk and per cell, so the cell-to-cell variance a bulk average hides is recoverable. Transcript and protein variation on one recombinant panel.
2	Hale 2024 (CRISPRi x natural variation)	Smith 2016 (CRISPRi chem-genetic) McGlinchy 2021 (genome-scale CRISPRi library) Bloom 2019 (16-parent cross)	CRISPR interference knockdown in S288C guide library design segregant genotype class, growth traits	One knockdown effect measured in the reference background and across 169 segregants, which separates a gene effect from a background effect. Whether a knockdown effect is a property of the gene or of the guide. Knockdown effect against natural allele effect on comparable panels.
3	Jackson 2020 (TF-deletion single-cell atlas)	Kemmeren 2014 Nadal-Ribelles 2025 (Perturb-seq) Airolidi 2016 (nitrogen-limited steady-state and dynamic transcriptome) Hackett 2020 (IDEA inducible-TF transcriptome time series)	single deletions of the same transcription factors, bulk against single cell deletion perturbation with a single-cell transcriptome readout the same nitrogen-limited media formulations the same transcription factors, deletion against induction	The same knockout read out as a population average and as a per-cell distribution, which is the only available calibration of what pseudobulk loses. Twelve genotypes over eleven conditions against about 3,500 genotypes over two, so genotype breadth and condition breadth are separable for once. A wild-type expression baseline in the same medium, which is what makes the deletion effect a contrast rather than an absolute. Loss of function against gain of function on one regulator set.
4	Puddu 2019 (WGS of the deletion collection)	Costanzo 2016 dmf Nadal-Ribelles 2025 (Perturb-seq)	the same deletion collection pooled deletion strains identified by barcode	Converts the sequence basis of every S288C-KO row from an assumption into a measurement, including the largest built datasets. Which pooled strains are not what their barcode says, which bears directly on the reported genotype-assignment impurity.
5	N’Guessan 2025 (segregant scRNA-seq eQTL)	Boocock 2025 (single-cell eQTL mapping) Bloom 2019 (16-parent cross)	segregant genotype class, single-cell transcriptome segregant genotype class, 16 founders against 2	Two independent single-cell expression quantitative trait loci maps, which is the replication neither has on its own. Whether an expression effect measured in a two-parent cross holds across sixteen founders.
6	Hackett 2020 (IDEA inducible-TF transcriptome time series)	Kemmeren 2014 Sameith 2015 dm	the same transcription factors, induction against deletion transcription-factor pairs	Whether a regulator’s targets are the same set whether it is removed or over-induced. Induction dynamics for the single factors whose double deletions carry a measured expression epistasis.
7	Hu 2007 (TF deletion expression compendium)	Kemmeren 2014	single deletions with a bulk expression readout	Two independently produced deletion expression compendia over overlapping regulators, which is the reproducibility check neither has alone and the largest such pair in yeast.

Table 3, continued

#	Candidate	Partner	Join key	What the join yields
8	Dong 2021 (MAGIC + SAM biosensor)	Jackson 2020 (TF-deletion single-cell atlas) Lian 2019 (MAGIC CRISPR-AID) Cachera 2023 (CRI-SPA betaxanthin)	transcription-factor deletions the same tri-functional library, fitness against biosensor readout metabolite biosensor or colorimetric product readout on a genome-scale library	269 regulators in bulk against 11 per cell, so the regulators the single-cell atlas cannot reach still carry a profile. One library, two labels: growth selection and product concentration, which is how a tolerance screen is converted into a production screen. Two product-labeled genome-scale screens on different chemistry.
9	Momen-Roknabadi 2020 (inducible CRISPRi library)	McGlincy 2021 (genome-scale CRISPRi library) SGD essentiality	independently designed guide libraries over the same genes essential genes, knockdown against deletion	The paired half of the library-transfer test. A graded phenotype where a deletion collection records only absence.
10	McGlincy 2021 (genome-scale CRISPRi library)	Momen-Roknabadi 2020 (inducible CRISPRi library) Smith 2016 (CRISPRi chem-genetic)	independently designed guide libraries over the same genes CRISPR interference over the same gene set	Library-to-library transfer, which is the cheapest decisive test of whether a model learned gene function or guide-design idiosyncrasy. A knockdown fitness prior for the chemical-genetic conditions already built.
11	Bao 2018 (CHANGE single-nucleotide library)	Lian 2019 (MAGIC CRISPR-AID) Li 2016 (tRNA fitness landscape)	guide-indexed genome-wide library from the same laboratory designed single-nucleotide variants with a fitness label	Sub-gene alleles against whole-gene activation, interference and deletion on one platform lineage. Genome-wide shallow variant coverage against single-gene exhaustive coverage.
12	Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	Lopez 2024 (isobutanol screen, private) Kuroda 2019 (isobutanol-specific tolerance)	isobutanol, knockdown against deletion isobutanol tolerance on the same collection	A graded knockdown axis over the phenotype the built biosensor screen measures as a titer proxy. Dose-graded knockdown against whole-gene deletion for one phenotype.
13	Roy 2018 (multiplexed precision editing)	Bao 2018 (CHANGE single-nucleotide library)	designed edits, single against multiplexed	Whether multiplexed edit effects are the sum of their single-edit effects.
14	Newman 2006 (protein noise, GFP library)	Messner 2023 (proteome) Gasch 2017 (single-cell stress heterogeneity)	protein abundance across the genome, per cell against population per-cell variation in an isogenic population	Which proteins have a population mean that no single cell is near, which decides where a mean-level model is the wrong object. Noise at the protein and transcript layers on the same question.
15	Mukherjee 2021 (CRISPRi essential genes x acetic acid)	Mormino 2022 (CRISPRi acetic-acid) Mota 2024 (weak-acid screen)	acetic acid, CRISPR interference weak-acid stress on deletion strains	The genome-scale extension of a twelve-gene screen already built. Essential-gene coverage for a phenotype the deletion collection can only sample from the non-essential side.
16	Lian 2017 (CRISPR-AID, beta-carotene)	Ozaydin 2013 (beta-carotene screen)	beta-carotene titer	Three perturbation modalities against a whole-collection deletion screen on the same product.
17	Hughes 2000 (compendium of expression profiles)	Kemmeren 2014 Hu 2007 (TF deletion expression compendium)	deletion strains, gene by gene, fourteen years and two array platforms apart deletion strains with a bulk expression readout	Cross-laboratory replication of the deletion transcriptome, which the built set cannot test against itself and which Nadal-Ribelles 2025 failed cross-batch. A third independent compendium, so agreement can be measured across three laboratories rather than asserted from two.

Table 3, continued

#	Candidate	Partner	Join key	What the join yields
18	Lenstra 2011 (chromatin regulator deletion expression)	Kemmeren 2014	single deletions with a bulk expression readout, different gene class	Chromatin regulators beside sequence-specific factors on one expression readout.
		Sameith 2015 dm	regulator deletions, single against double	Whether chromatin-regulator effects combine the way transcription-factor effects do.
19	Sun 2013 (mRNA synthesis and decay rates across deletion strains)	Kemmeren 2014	single deletion strains, gene by gene	A steady-state transcript level beside the synthesis and decay rates that produce it, so two strains with the same level but different kinetics stop looking identical.
		Martin-Perez 2017 (protein half-lives)	turnover, transcript side against protein side, gene by gene	Both half-life terms in one place, which is what an RNA-to-protein model needs in order to have a reason for the two to disagree.
20	Jariani 2020 (yeast scRNA-seq through lag phase)	Jackson 2020 (TF-deletion single-cell atlas)	carbon-source shift with a per-cell readout	A transition sampled densely in time where the atlas samples eleven steady states.
21	Guo 2018 (CRISPR-Cas9 tiling of essential genes)	SGD essentiality	essential genes	A graded allelic series where the built record is binary.
22	Urbonaite 2021 (yeastDrop-Seq under drug treatment)	Hoepfner 2014 (HIP/HOP atlas)	chemical treatment of yeast	A per-cell readout for compound response where the built atlas has a pooled fitness score.
23	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	Nadal-Ribelles 2025 (Perturb-seq)	same laboratory, protocol against application	The sensitivity ceiling of the platform the built genome-scale single-cell dataset was produced on.
		Gasch 2017 (single-cell stress heterogeneity)	isogenic per-cell variation in BY4741	Two independent measurements of how much an unperturbed yeast population varies, which is the floor a perturbation must clear.
24	Gasch 2017 (single-cell stress heterogeneity)	Gasch 2000 (environmental stress response)	the same stress conditions, bulk against single cell	How much of the canonical stress response is a population average of cells that are individually in different states.
25	Jaffe 2019 (multiplexed CRISPR interference epistasis)	Costanzo 2016 dmi	gene pairs, knockdown against deletion	Whether digenic interaction measured between two nulls is recovered between two knockdowns, which decides if the built interaction data can supervise a knockdown campaign.
		Kuzmin 2018 tmi	higher-order gene combinations	The knockdown analog of trigenic interaction.
26	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	Jackson 2020 (TF-deletion single-cell atlas)	same platform and laboratory, wild type against deletions	The unperturbed per-cell variance the perturbed atlas is read against.
27	Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	Jackson 2020 (TF-deletion single-cell atlas)	multiplexed genotypes per single-cell run	The throughput ceiling for a barcoded-genotype design, measured rather than assumed.
28	Mulleder 2012 (prototrophic deletion collection)	Mulleder 2016 (amino-acid metabolome)	the same prototrophic deletion strains	The strain resource the built amino-acid metabolome was measured on, which is what makes defined-medium phenotyping interpretable.
		Jackson 2020 (TF-deletion single-cell atlas)	prototrophy as a precondition for minimal media	A ready-made prototrophic library for a campaign that needs nitrogen or carbon limitation.
29	Su 2023 (single-cell transcriptomes under four stresses)	Gasch 2000 (environmental stress response)	osmotic and starvation stress	Full-length per-cell transcripts against the bulk stress-response reference.
30	Wang 2022 (single-cell transcriptomes across replicative aging)	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	wild-type per-cell states over time	Replicative age as a covariate on a platform that does not measure it.

Table 3, continued

#	Candidate	Partner	Join key	What the join yields
31	Albert 2018 (eQTL in 1,012 segregants)	Jakobson 2025 (genome-to-proteome map) Gerke 2017 (urea-cycle mQTL)	the same segregant panel segregant genotype class	The transcript half of a paired transcript and protein map. A metabolite locus read through the transcripts of the pathway that produces it.
32	Jakobson 2025 (genome-to-proteome map)	Albert 2018 (eQTL in 1,012 segregants) Messner 2023 (proteome)	the same sequenced segregant panel protein abundance, natural variation against deletion	Protein and transcript quantitative trait loci on one genotype set, which is the direct measurement of how much of protein variation transcript variation explains. Whether the proteome response to a deleted gene resembles the response to a natural allele of it.
33	Muenzner 2024 (natural-isolate proteome)	Caudal 2024 (pan-transcriptome) Dutta 2026 (barcoded natural-isolate chemical response) Teyssonniere 2024 (species-wide proteome against transcriptome)	the sequenced 1,011-isolate panel the same 1,011-isolate panel the 1,011-isolate panel, protein layer	Transcriptome and proteome on overlapping isolates, which is the cheap-layer against expensive-layer transfer question at natural genetic distance. Genome, transcriptome, proteome and a chemogenomic response surface on one genotype axis. De-duplication before either is built: 796 isolates here against 942 there, shared authors and one panel, and no evidence yet on whether the two acquisitions are independent.
34	Aulakh 2025 (genome-scale ionome)	Mulleder 2016 (amino-acid metabolome) Kemmeren 2014	the whole deletion collection, metabolite readout the same deletion collection	Two orthogonal metabolic panels on one genotype axis. Element levels against the transcriptional response of the same knockout.
35	Teyssonniere 2024 (species-wide proteome against transcriptome)	Caudal 2024 (pan-transcriptome) Muenzner 2024 (natural-isolate proteome)	the same sequenced isolates, 942 against 943 the 1,011-isolate panel, protein layer	Transcript and protein for the same strain, one to one rather than by overlap. The published result on this pair is that the two correlate weakly and share 3 percent of their associated variants, so it is the calibration an inference model is scored against. De-duplication before either is built: the two share authors and a panel, and whether they are independent acquisitions is unsettled.
36	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	Albert 2018 (eQTL in 1,012 segregants) Jakobson 2025 (genome-to-proteome map) Leutert 2023 (phosphoproteome x 101 conditions)	BY x RM segregants, transcript layer segregant genotype class, protein layer phosphosite identity	112 strains with transcript and protein measured together against 1,012 with transcript alone, so the within-strain RNA-to-protein residual fitted on the small panel can be applied to the large one. Two segregant proteomes on different panels and platforms, which is what separates a protein quantitative trait locus from a batch. Phosphorylation driven by genotype against phosphorylation driven by condition, on one site vocabulary.
37	Leutert 2023 (phosphoproteome x 101 conditions)	Gasch 2000 (environmental stress response) Messner 2023 (proteome)	overlapping stress conditions protein identity	Post-translational regulation against transcriptional regulation for one condition set, on the layer that acts first. Modification state against abundance, which abundance alone cannot separate.
38	Brauer 2008 (growth-rate-controlled chemostat transcriptome)	Boer 2010 (metabolome across nutrient limitations) Airolidi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	the same chemostat limitations growth rate under limitation, carbon against nitrogen	The transcript half of a paired transcript and metabolite series. Whether the growth-rate expression program is nutrient-general.

Table 3, continued

#	Candidate	Partner	Join key	What the join yields
39	Hackett 2016 (SIMMER multi-omic flux)	Boer 2010 (metabolome across nutrient limitations) Brauer 2008 (growth-rate-controlled chemostat transcriptome)	the same nutrient-limited chemostats nutrient limitation at controlled growth rate	Metabolite concentration and flux under one condition set, which is what a kinetic model needs and neither supplies alone. Expression, metabolite and flux over one condition axis.
40	Zhu 2014 (kinase / phosphatase lipidomics)	da Silva 2014 (lipidomics) Kemmeren 2014 Xue 2025 (free fatty acids, private)	lipid species on deletion strains kinase and phosphatase deletions fatty-acid metabolism	A second lipidome on a signaling-focused genotype set. Lipid composition against the transcriptional response of the same signaling mutant. Lipid class distribution against free fatty-acid titer, the two sides of the malonyl-CoA sink.
41	Gerke 2017 (urea-cycle mQTL)	Albert 2018 (eQTL in 1,012 segregants) Mulleder 2016 (amino-acid metabolome)	segregant genotype class nitrogen and amino-acid metabolites	A metabolite locus and the expression loci in the same interval. Natural-allele against deletion effects on one metabolic module.
42	Ambroset 2014 (metabolite QTL)	Peeters 2021 (fermentation-trait QTL atlas) Zelezniak 2018 (metabolome)	segregant panels with fermentation traits metabolite panel	74 metabolites against 18 mapped traits, so a trait locus can be read as a metabolite change. Recombinant against deletion genotypes on a metabolite readout.
43	Blank 2005 (13C metabolic flux)	Kemmeren 2014 Mulleder 2016 (amino-acid metabolome)	deletion strains present in both deletion strains, flux against pool size	Flux against expression for the same knockout, which is the only place the transcript-to-flux step can be fitted rather than assumed. Whether a changed pool reflects changed flux or changed demand.
44	Skelly 2013 (expression variation across isolates)	Muenzner 2024 (natural-isolate proteome)	natural isolates, paired against joined modalities	A within-experiment paired transcript and protein measurement to calibrate the cross-study join.
45	Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	Jackson 2020 (TF-deletion single-cell atlas) Yu 2021 (proteome and metabolome under nitrogen limitation)	the same nitrogen-limited media nitrogen limitation	A wild-type bulk baseline for the medium the single-cell atlas perturbs in. Transcript, protein and metabolite under one limitation.
46	McManus 2014 (ribosome profiling, allele-specific)	Messner 2023 (proteome) Martin-Perez 2017 (protein half-lives)	gene identity, translation rate against protein abundance gene identity, synthesis against degradation	A per-gene translation term to put against measured protein level, which is the coefficient a model otherwise has to learn blind. Both halves of protein turnover, so steady-state abundance can be decomposed rather than only predicted.
47	Martin-Perez 2017 (protein half-lives)	Messner 2023 (proteome) Kemmeren 2014	gene identity, half-life against abundance gene identity, protein half-life against transcript response	Which proteins are abundant because they are made fast and which because they are destroyed slowly, a distinction abundance alone cannot make. Whether a transcript change reaches the protein layer at all, which for a long-lived protein it largely does not.
48	Boer 2010 (metabolome across nutrient limitations)	Brauer 2008 (growth-rate-controlled chemostat transcriptome)	the same chemostat limitations and growth rates	The metabolite and transcript halves of one experimental series.

Table 3, continued

#	Candidate	Partner	Join key	What the join yields
		Zelezniak 2018 (metabolome)	metabolite panel	Condition-driven against genotype-driven metabolite variation.
49	Tengolics 2024 (domestication metabolome)	Caudal 2024 (pan-transcriptome)	sequenced natural isolates	Metabolite against transcript variation across the same population.
50	Eder 2020 (flux QTL)	Blank 2005 (13C metabolic flux)	flux measurement, natural variation against deletion	Whether flux control points found by deletion are the loci natural variation actually moves.
		Albert 2018 (eQTL in 1,012 segregants)	segregant genotype class	Flux loci against expression loci on comparable panels.
51	Foss 2007 (BY x RM segregant proteome)	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	the same cross, protein layer, fifteen years and two platforms apart	Cross-laboratory replication of segregant protein abundance, which neither measurement can establish alone.
52	Yu 2021 (proteome and metabolome under nitrogen limitation)	Airoidi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	nitrogen limitation	The transcript layer for the protein and metabolite layers measured here.
53	de Boer 2020 (100M random promoters)	Vaishnav 2022 (regulatory DNA fitness landscape) Renganaath 2020 (natural promoter-variant MPRA)	the same random promoter library, expression against fitness promoter sequence with a measured expression readout	Two labels on one sequence set, which turns a sequence-to-expression model into a sequence-to-fitness model without new data. Whether a model trained on random sequence predicts the effect of a real allele, the cheapest decisive transfer test in the table.
54	Vaishnav 2022 (regulatory DNA fitness landscape)	de Boer 2020 (100M random promoters)	the same promoter library	Fitness label for sequences that already carry an expression label.
55	Lee 2014 (HIP-HOP fitness signatures)	Hoepfner 2014 (HIP/HOP atlas)	the same heterozygous and homozygous collections, compound response	Two chemical-genetic matrices on one genotype axis, which is enough compound overlap to measure cross-screen reproducibility directly.
		Hillenmeyer 2008 het (FitDb HIP)	the same collections and readout	A third independent screen of the same design.
56	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	Bloom 2019 (16-parent cross)	segregant genotype class, panel size	One cross at 100,000 progeny against sixteen founders at 14,000, which separates panel size from allelic diversity.
57	Galardini 2019 (four backgrounds x 38 conditions)	Costanzo 2021 (condition-SGA) Hale 2024 (CRISPRi x natural variation)	deletion strains across conditions perturbation held fixed, background varied	The same conditional fitness question asked in four backgrounds rather than one. Deletion against knockdown for the background-dependence question.
58	Parsons 2006 (bioactive-compound profiling)	Hoepfner 2014 (HIP/HOP atlas)	compound response on deletion strains	Compound overlap across screens run on different platforms.
59	Dutta 2026 (barcoded natural-isolate chemical response)	Caudal 2024 (pan-transcriptome)	the sequenced 1,011-isolate panel	Chemical response and transcriptome on one isolate set.
		Hoepfner 2014 (HIP/HOP atlas)	compound response, isolates against deletions	Whether compound sensitivity found by gene deletion predicts sensitivity across natural genetic backgrounds.
60	Li 2016 (tRNA fitness landscape)	Domingo 2018 (tRNA double-mutant landscape)	the same tRNA gene, single against double mutants	Whether a single-mutant landscape predicts the double-mutant one, which is the within-gene form of the epistasis question.

4 What was dropped x

Table 4: Considered and dropped. *no-sequence* is the hard gate: without a route to the strain’s genomic content there is no genotype to map a phenotype from.

Dataset or group	Rule	Reason
Snoek 2015 robot-assisted genome shuffling (ethanol tolerance)	no-sequence	Shuffled progeny are recombinants of unrecorded parental segments with no per-strain sequencing, so the genotype cannot be reconstructed.
Adaptive laboratory evolution / EMS mutagenesis tolerance panels (e.g. farnesol-FPP selection)	no-sequence	Mutations are unmapped, so a strain’s genome is unknown even in principle without whole-genome resequencing the study did not do.
Avalos lab isobutanol-biosensor deletion screen	not-a-dataset	Described only in a PhD thesis and a DOE report. Track for publication; not citable as a dataset.
Kitamoto/Kaneko ester and higher-alcohol screen	not-a-dataset	Citation could not be pinned to a specific paper. A placeholder, not a located dataset.
Fuhrer/Zamboni flow-injection metabolomics	off-species	The validated genome-scale application is in <i>E. coli</i> . No yeast deletion-collection version was located.
Anglada-Giroto 2022 (CRISPRi + metabolomics)	off-species	352 CRISPRi genes against 1,342 drug-induced metabolic changes, but in <i>E. coli</i> . Verified from the mirrored PDF.
Wildenhain 2016 (Sci Data chemical-genetic matrix)	already-built	A data descriptor, not a second experiment. Its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly what torchcell/datasets/scerevisiae/wildenhein2015.py already ingests: 242 strains, 5,518 compounds, 492,126 interaction tests.
O’Duibhir 2014 expression compendium	already-built	The paper’s expression data IS the Kemmeren 2014 compendium plus PCA transforms, already built, as oduibhir2014.py states; only its growth-rate readout was new and that is built too. Caught by check_candidate_overlap.py rather than by inspection, which is the first time the script found a duplicate before a person did.
Ozturk 2022 proteome-effects screen	off-species	3,308 deletions in <i>S. pombe</i> . Out of species scope.
Chen 2024 <i>K. phaffii</i> CRISPR fitness screen; Y1000+ multi-species panel	off-species	Not <i>S. cerevisiae</i> . Revisit if the generalization axis widens to other hosts.
iIsor850 genome-scale model (<i>Issatchenkia orientalis</i>)	not-a-dataset	A reconstruction, not an experiment, and off species. No genome-wide <i>I. orientalis</i> phenotype screen was located; the CABBI knockout library is unpublished.
CABBI 13C-MFA kinetic-model repository	not-a-dataset	16 knockout strains total, and the released fluxes are K-FIT model fits recapitulating 75-77 percent of measurements, not raw measurement. Blank 2005 supersedes it on both counts.
YEASTRACT+ (Teixeira 2023; Monteiro 2020)	not-a-dataset	A literature-curated knowledge base of 304,547 documented TF-target associations, each tagged with the evidence type (DNA binding or expression) and with the environmental condition in which it was observed, so a condition-scoped regulatory graph is recoverable from it. It is a prior over the genome, not a genotype-by-environment measurement, and it belongs in the graph layer beside GO and the metabolic model rather than in this table. Both papers are mirrored; the flat files are shared on request by the maintainers and are not yet held, and scYeast (Fan 2027) built its attention prior from the documented-targets subset (about 170,000 pairs).
Sc2.0 / Sc3.0 synthetic-genome design	not-a-dataset	Genome-engineering programs, not gene-indexed perturbation-phenotype data.
Lee 2014 / Hoepfner 2014 / Vanacloig-Pedros 2022 / Messner 2023 / Mulleder 2016 / Lian 2019 / Mormino 2022 / Nadal-Ribelles 2025 and 12 others	already-built	Named as top candidates by the 2026-07 triage pass and BUILT since. Only Lee 2014 remains outstanding and is kept as row 4.
Bloom 2019 (16-parent cross)	already-built	Ranked 12 in the previous pass and built since as the 50th supported dataset: 13,950 sequenced segregants over 38 traits, L0-L4 verified. Its place in this pass is as a join partner, not a candidate.

Dataset or group	Rule	Reason
TF-target prior networks used by Jackson 2020 (YEASTRACT, ATAC-motif of Castro 2019, Bussemaker affinity of Ward 2008, Tchourine 2018 gold standard)	not-a-dataset	Four regulator-by-gene matrices: 11,486 unsigned YEASTRACT edges over 3,912 genes and 152 TFs, 71,865 signed ATAC-motif edges over 5,551 genes and 138 TFs, a dense 6,516 by 123 affinity matrix, and the 1,403-edge signed gold standard. Each is a prior over the genome with no genotype and no environment, so each belongs in the graph layer beside GO and YEASTRACT+, not in a genotype-by-environment table. All four ship as TSVs in the eLife Source code 1 archive.
Tchourine 2018 bulk expression compendium (2,577 observations)	not-a-dataset	The benchmark Jackson 2020 compares its single-cell network against. It is a re-aggregation of public bulk series, and the strain behind an observation lives in each source series' metadata rather than in the released matrix, so admitting it is a de-duplication and re-curation task against Kemmeren 2014, Gasch 2000 and Brauer 2008 rather than one loader. Revisit if the per-sample genotype table is located.
Albert 2014 (X-pQTL, single-cell protein abundance in large populations)	not-a-dataset	Albert FW, Treusch S, Shockley AH, Bloom JS, Kruglyak L. Nature 2014;506:494-497. Considered as the BY x RM segregant proteome and it is not one: protein level is read from a green fluorescent protein fusion for one gene at a time in a large unsequenced pool, and the mapping comes from sorting that pool rather than from genotyped individuals. There is no strain-by-protein matrix to store, so it cannot enter as a proteome row. The segregant proteome rows here are Grossbach 2022 and Foss 2007.
Scholes 2019 bulk RNA-seq control (GEO GSE135430)	not-a-dataset	Jackson 2020's external bulk reference point. One wild-type BY4741 genotype in one condition, and the study's own variable is the RNA isolation protocol, so there is neither a genotype nor an environment axis to record.
Brandner 2025 (mapSPLiT, CRISPRa and CRISPRi single-cell transcriptomes in <i>E. coli</i> and <i>P. putida</i>)	not-a-dataset	The only microbial dataset found that reaches both Perturb-seq axes on purpose: 52 transcription factors targeted by 118 guides, pooled, with multi-guide combinations included to resolve genetic interactions, read out as a single-cell transcriptome. <i>Pseudomonas putida</i> is a single-cell bioproduction host and so is in scope for the generalization axis. It is a preprint and no deposited accession was found, so there is no scriptable data route and it is excluded rather than ranked. It is the closest existing design template for the campaign and should be revisited on publication.
Mammalian Perturb-seq (Replogle 2022, Zhu 2026, Dixit 2016, Datlinger 2017, Yao 2024)	off-species	The format references, and off species. Replogle perturbs every expressed gene by CRISPR interference across more than 2.5 million cells; Zhu 2026 crosses a genome-scale perturbation with several stimulation time points across about 22 million cells, which is the perturbation-by-context tensor a yeast campaign is an analog of; Yao 2024 pools random multi-perturbation composites over 598 genes and decompresses them, which is the one published route to second-order genetic interactions at a tractable cell count. None supplies yeast data; all four inform the design.
Bacterial single-cell transcriptomics (Wang 2023 M3-seq, Ma 2023 BacDrop, McNulty 2023 ProBac-seq, Kuchina 2021 microSPLiT)	off-species	Off species, and the axis is environmental rather than genetic: hundreds of thousands of cells over stress, antibiotic and growth-stage conditions in isogenic backgrounds. No perturbation-by-readout matrix to ingest.
Fulcher 2024 (nanoSPLITS), Taniguchi 2010, Baronas 2026, Leonaviciene 2023 and 2020, Datlinger 2021, Macosko 2015, Rosenberg 2018	off-species	Platform and paired-modality papers, all off species. nanoSPLITS measures a transcriptome and a mass-spectrometry proteome from the same single cell, which is the precedent for inferring an expensive layer from a cheap one on paired anchors; Taniguchi 2010 is the reference for protein and mRNA copy-number noise in a microbial host. Neither releases a perturbation matrix.
Reviews, protocols and single-cell statistics in the collection (Nadal-Ribelles 2024, Sun 2023, Larson 2013, Zun 2023, Gaisser 2024, Squair 2021, Zhang 2020, Svensson 2020, Grun 2014, Robinson 2008, McCarthy 2012, Hart 2013, Wang 2026)	not-a-dataset	Thirteen items with no perturbation data of their own. They are the sizing and false-discovery discipline for the campaign, not rows: pseudobulk replicate variation, sequencing depth per cell, dispersion estimation, and whether droplet counts are zero-inflated.

The **already-built** row in that table is the largest single correction this pass makes. A July 2026 triage pass named a top ten for immediate ingestion; nine of those ten have since been built, along with roughly a dozen other rows it

ranked highly. Only Lee 2014 remains outstanding, and it is row 56 here. Bloom 2019 joined them since the previous pass and is the 50th supported dataset. Any list that still treats that top ten as outstanding work is stale.

4.1 Four rows were not what their names said ✗

Checking a name against the built list is not sufficient. A dataset can carry a loader, a failed retrieval attempt, or an already-ingested deposit under a different paper's name. Three rows did one of those each, and a fourth was caught by the script in Sec. 4.2: O'Duibhir 2014's expression data is the Kemmeren compendium plus PCA transforms, already built, as that loader's own docstring states. It is the first duplicate here found mechanically rather than by a reader, which is the point of the check.

Wildenhain 2016 was already built, under the 2015 name. It is a data descriptor rather than a second experiment, and its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly the accession `torchcell/datasets/scerevisiae/wildenhain2015.py` already ingests. The identity is not a judgment call: the 2016 paper reports 242 sentinel strains, 5,518 compounds and 492,126 interaction tests, and the loader's own docstring records 242 systematic ORFs and 492,126 datapoints from that AID. The row is removed.

That leaves a citation-fidelity defect in the built dataset worth fixing separately. The data it ingests is the *extended* matrix described by the 2016 release, but it is filed under the 2015 paper, which reports a strictly smaller experiment: 195 sentinels and 4,915 compounds. The loader already half-flags this, noting that the released data covers 242 ORFs rather than the 195-strain panel it quotes. Under the sourcing rule that a value carries the citation it was actually read from, the released matrix should cite the 2016 descriptor, with 2015 as the original subset.

Lee 2014, row 56, is blocked rather than available. WS15 already attempted this ingestion. The roadmap records it as awaiting author matrices, which means the Science supplementary tables and the Nislow and Giaever lab portal did not yield per-strain values. Its rank is still earned: 3,356 compounds against the full heterozygous and homozygous collections really is the largest chemical-genetic matrix in existence, and it really is on the YKO genotype axis the built set already uses. What changes is the nature of the work. Row 56 is an author request, not a loader, and the 1.3×10^7 instances it contributes to the totals in Sec. 6 should not be counted as reachable until that request succeeds.

Turco 2023, row 71 of the full ranking and outside Table 1, already has a loader. `torchcell/datasets/scerevisiae/yeastphenome.py` consumes the curated YeastPhenome release and retains 49 growth screens, so this row is in flight rather than net new. It is not in the built set, which is why it appears here at all. Two consequences follow. Its de-duplication problem is partly solved already, since the loader excludes the primaries built directly, along with heterozygous-diploid and expression screens. And Lee 2014 is not among its 49 screens, so ingesting YeastPhenome does not backfill row 56.

4.2 Overlap is now checked by script, and the check is not yet strong ✗

Three rows turned out to be already held and none was caught by inspection, so the check is now `experiments/database/scripts/check_candidate_overlap.py`, run against the candidate JSON. It applies three keys of increasing strength: the candidate DOI cited in a built loader, an accession token shared with one, and the candidate PMID appearing in an aggregator's source-PMID column.

It currently reports one overlap, Turco 2023, which Sec. 4.1 already records as a loader in flight; a second, O'Duibhir 2014's expression compendium, was found and removed during this pass. It reports no overlap with SynLethDB across 3,307 source PMIDs spanning both the lethality and rescue layers. That covers the question of whether the genetic-interaction rows are already re-served: Sharifpoor 2012, Decourty 2021 and Braberg 2020 are all absent. For Sharifpoor the absence is structural rather than a curation gap, since SynLethDB models synthetic lethality and synthetic rescue, both two-loss-of-function events, and synthetic dosage lethality pairs an overexpression with a deletion.

The honest limit is coverage, and the expansion made it worse in relative terms. The script resolves a PMID for 56 of 168 rows and finds a parseable accession in 9. Accession is the strongest key and the one that would have caught Wildenhain 2016, so it is running on about one row in seventeen. Recording each row's deposit identifier is the single highest-value edit available to this list, and it now matters more than it did at 73 rows. It is also the check that would have caught Bloom 2019 leaving the list a pass earlier than a reader did.

4.3 The expansion tranche is less checked than the first 73 ✗

Rows carry a confidence value in the generated JSON. *sourced* means the figures trace to something read this session or to the sourced triage note, and covers 75 of the 168 ranked rows, 34 of the sixty printed. *recall* means the row was

written from domain knowledge, and covers 93 of the 168, 26 of the sixty. For a recall row the dataset is real and the description is sound, but the citation details, the exact counts and the accession all need confirming before a loader is written, and a few carry an explicit warning in the accession column where a collision or a species question is open.

The two tranches are not interleaved by accident: ranking put several recall rows at the very top, because de Boer 2020 and Boocock 2025 genuinely are the largest things here. That is the right ranking and the wrong reading if the confidence flag is ignored. Treat the top of the table as a claim to verify first, not a queue to start work from.

Banding does not change that and in one place sharpens it. Hackett 2020 at rank 6, Hu 2007 at 7, Hughes 2000 at 17, Lenstra 2011 at 18 and Sun 2013 at 19 are all *recall* rows, so five of the first twenty carry accessions that have not been confirmed against a deposit.

The ten rows added over the last two passes are a tranche of their own and are weaker again. Their citations, venues and headline counts were verified against PubMed, which is why the author lists and page ranges can be relied on. Their deposits were not: no accession among them was fetched, and none of the ten papers is in the mirror, so none has been read beyond its abstract. Foss 2007 is the extreme case and is recorded as such, with neither a segregant count nor a proteome depth. Teyssonniere 2024 carries a second warning that is not about confidence: it shares authors and an isolate panel with Muenzner 2024, and whether the two proteomes are independent acquisitions has not been established, so building both without settling that would risk ingesting one measurement twice.

5 Before a loader is written ✗

Every figure in Table 1 is an axis count, not a verified record count, and several are weaker than that. Three qualifications apply, in decreasing order of how much damage they would do if ignored.

Instance counts marked † are products, not counts. A screen reported as 4,800 strains against 3 conditions is entered as 14,400 instances. Real screens drop strains per condition, so each such figure is an upper bound. The ones marked ‡ are weaker still: order-of-magnitude estimates for libraries whose exact size was not confirmed this pass. Neither should be quoted as a dataset size without opening the source.

Phenotype dimensionality is estimated where no fixed panel is published. Ranking uses instances times dimensionality, and dimensionality is exact for a stated panel such as Ambroset 2014 at 74 metabolites, but estimated for open-ended ones. Proteome rows are entered at 2,000 proteins per strain and transcriptome rows at 6,000 genes. Those estimates move rank order, so a row whose position looks surprising should have its dimensionality checked first.

Accessions marked unconfirmed were not fetched. The accession column separates locations verified live this pass from those taken from a paper’s data-availability statement. Confirmed live: GEO GSE123118 with its companion GitHub release for Galardini 2019, SRA PRJNA986287 for Hale 2024, PRIDE PXD048219 for Muenzner 2024, Dryad for Chica 2026, ChemGRID for Wildenhain 2016, and BioProject PRJNA379146 for Cubillos 2017. Everything else is a claim from a publication, and the pattern is well established that a data-availability statement and a reachable file are different things.

Four rows carry a risk specific enough to name. Triikka 2015 is recorded as figure-only, so whether Additional file 1 holds per-strain scores decides whether Triikka is ingestible at all. Chang 2012 and the Gurvitz fatty-acid screen are PDF-table-only and need manual transcription. Sugiyama is a non-peer-reviewed technical report and has the lowest provenance confidence of anything listed; it belongs in the reserve as a source of candidate genes rather than as training data.

Two rows are not single datasets. Turco 2023 is a meta-aggregation and Zackrisson 2016 is a platform. Neither reduces to one loader: Turco needs a de-duplication pass against the rest of this table before its net-new content is known, and Scan-o-matic needs a specific released condition panel chosen before it becomes a row rather than a plan.

6 Summary statistics ✗

Table 5: Summary of the final sixty, split at the fifty line. *Top 50* and *Extra* count rows; *Genotypes*, *Instances* and *Meas.* sum the row axes over all sixty and are lower bounds, since a row with no count contributes nothing. The sixty rows name 109 joins, 56 of them to a dataset already built.

	Top 50	Extra	Genotypes	Instances	Meas.
<i>By band</i>					
perturb-seq	30	0	628,459	1.8×10^6	3.8×10^8
molecular layers	20	2	9,187	9.4×10^3	1.3×10^7
scale	0	8	120,196,664	1.4×10^8	1.4×10^8
<i>By tier</i>					
tier 1	8	8	120,500,126	1.4×10^8	4.8×10^8
tier 2	21	0	287,496	3.6×10^5	4.2×10^7
tier 3	19	1	46,685	4.1×10^4	4.1×10^6
tier 4	2	1	3	17	5.6×10^4
<i>By class</i>					
CRISPR library screen	8	0	307,294	3.7×10^5	3.7×10^5
Deep mutational scan	0	1	65,000	6.5×10^4	6.5×10^4
Expression / single cell	20	0	5,896	6.3×10^4	3.8×10^8
Metabolite / precursor	4	1	5,131	5.2×10^3	1.1×10^5
Modality / backbone	5	0	13,328	9.8×10^3	5.1×10^4
Natural variation	11	4	410,561	3.1×10^6	1.5×10^7
Regulatory DNA	0	2	120,000,000	1.2×10^8	1.2×10^8
Tolerance / robustness	2	2	27,100	1.3×10^7	1.3×10^7
<i>By sequence basis</i>					
S288C+designed-edit	3	1	141,200	1.4×10^5	9.7×10^6
S288C+guide	7	0	242,194	3.1×10^5	3.1×10^5
S288C+reporter-locus	0	2	120,000,000	1.2×10^8	1.2×10^8
S288C+tag	1	0	2,500	5.0×10^3	1.0×10^4
S288C-KO	10	2	32,745	1.3×10^7	2.0×10^7
engineered-chassis	1	0	200	200	200
isolate-WGS	4	2	17,441	8.9×10^5	4.5×10^6
reference-only	15	1	17	243	1.1×10^6
segregant-WGS	9	2	398,013	2.2×10^6	3.7×10^8
<i>Perturb-seq axis</i>					
high on both axes	3	0	5,905	6.1×10^4	3.7×10^8
input axis only	4	4	120,556,943	1.2×10^8	1.2×10^8
output axis only	11	0	3,151	7.2×10^3	1.3×10^7
neither axis	32	6	268,311	1.5×10^7	2.5×10^7
<i>Other attributes</i>					
carries a time axis	7	0	251	1.7×10^3	1.1×10^7
has a join to a built dataset	36	5	752,055	1.7×10^7	4.0×10^8
has a join to another candidate only	14	5	120,082,255	1.2×10^8	1.3×10^8
figures sourced this pass	29	4	623,730	1.6×10^7	3.8×10^8
figures from recall, to confirm	21	6	120,210,580	1.2×10^8	1.5×10^8
blocked on data access	0	1	11,000	1.3×10^7	1.3×10^7
instances a reported count	28	2	100,091,063	1.1×10^8	1.6×10^8
instances a product of the axes	6	4	415,718	3.5×10^6	3.5×10^8
instances an estimate	16	4	20,327,529	2.0×10^7	2.1×10^7
Total	50	10	120,834,310	1.4×10^8	5.3×10^8

The fifty are the two priority bands in full: 30 perturb-seq rows and 20 molecular-layers rows, with no scale row among them. The ten extra are the three molecular-layers rows the fifty line cut and the first seven scale rows, and those seven include the two largest datasets in the whole ranking, de Boer 2020 and Vaishnav 2022, which is why the instance and measurement sums in Table 5 are carried mostly by the extra ten rather than by the fifty. The fifty are vector-valued: a few hundred strains each carrying a transcriptome, a proteome or a per-cell distribution, so they hold few instances and many measurements, and ranking on instances alone would have ordered the two blocks the other way round.

Two class facts follow from the bands. Expression and single-cell rows are the largest class in the sixty, and 16 of the sixty are **reference-only**, meaning the environment rather than a strain edit carries the perturbation, which is what a condition-crossed expression layer looks like. Thirteen of the sixty sit on the S288C deletion collection, the axis most of the built set already uses, so those thirteen are the cheapest joins in the table.

6.1 Time ✗

A review comment on the previous version, anchored at Hackett 2020’s time points, raised the concern that nothing in the substrate carries a time component, and that one is needed eventually for concentration-over-time modeling of the ordinary-differential-equation kind. Both halves are right. Seven of the sixty rows carry a time dimension (Table 1, *Time* column): Hackett 2020’s induction series, Jariani 2020’s lag-phase course, Jackson 2023’s continuous course, Wang 2022’s three ages, Airolidi 2016’s upshift course, and the two rate rows, Sun 2013 and Martin-Perez 2017, whose rates are derived from a labeling series that is not itself released. The steady-state chemostat rows (Brauer 2008, Boer 2010, Hackett 2016, Yu 2021) vary dilution rate, which is a growth rate, not a time axis, and are not counted.

What the schema holds today is `Environment.duration_hours` and `Environment.duration_generations`, treatment durations that make two exposure lengths two distinct environments. A time series can therefore be ingested now, as one environment per time point, and that is how Hackett 2020’s roughly 200 records per strain are counted in Table 1. It is a flattening: a model sees time as a categorical condition, and nothing on the record says that eight environments are one series in order. A first-class time field on the record and a series identity that groups its points are the two schema additions a $c(t)$ model needs, and the seven rows above are the ones that would exercise them. They are recorded here as an open schema item rather than folded into this list’s ranking, since the ranking rule does not yet score a time axis.

6.2 Sun 2013 against the other deletion-strain expression sets ✗

A second comment asked whether Sun 2013 (row 19) overlaps the other deletion-strain expression datasets. The overlap is on the strain axis and not on the measurement. The built Kemmeren 2014 (1,484 deletions) and the candidates Hughes 2000, Hu 2007 and Lenstra 2011 each store a steady-state transcript level per gene and per deletion; Sun 2013 stores the synthesis rate and the decay rate that produce that level, across 46 deletions of mRNA degradation and metabolism genes, and none of the others stores a rate. It is therefore a second layer on shared strains, the join named in Table 3, rather than a second copy of a layer already held. What is not settled is the exact shared-strain count: the 46 are single deletions expected to be inside the Kemmeren set by gene identity, but the GEO strain list was not fetched, and the abstract does not say the strains came from the deletion collection. The row’s figures are marked *recall* for that reason.